When Words Save Watts: Government Communication and Household Electricity Use

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Abstract

A fundamental question for policymakers is whether communication can serve as an effective policy instrument. This paper examines France's energy crisis of 2022, during which surging prices prompted one of Europe's largest conservation campaigns. The present study investigates the relationship between official messaging, public attention and electricity demand. The analysis of the causal chain reveals that the effectiveness of communication depends on two factors: message framing and household flexibility. The findings indicate that information campaigns have the capacity to augment existing demand-side flexibility rather than creating it, thereby constraining their effectiveness as a standalone policy instrument.

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1 Introduction

A central challenge in economic policy is how to mobilise households to adjust demand when supply-side flexibility is exhausted. Traditional instruments, such as prices, regulations, or subsidies for new technologies, often operate with delay and may generate inefficiencies, for instance through deadweight losses arising from imperfect taxation schemes. By contrast, communication and informational appeals can be deployed rapidly and at low fiscal cost, offering policymakers an additional lever when time and resources are constrained. The effectiveness of such instruments to adjust household energy consumption, however, remains uncertain.

While theoretical frameworks suggest conservation incentives reduce informational internalities (List et al., 2023; Allcott and Mullainathan, 2010; De Young, 2000; Hungerford and Volk, 1990), with crisis framing amplifying salience (Curotto et al., 2025; Spence et al., 2021; Kudesia and Lang, 2024; Dyer and Kolic, 2020; Scheufele and Tewksbury, 2007), meta-analyses reveal heterogeneous empirical impacts (Abrahamse et al., 2005; Andor and Fels, 2018; Blasco and Gangl, 2023; Delmas et al., 2013; Brandon et al., 2017), comparative feedback yields the strongest effects, though realistic expectations remain modest and decay without reinforcement (Buckley, 2020; Allcott and Rogers, 2014). Whether large-scale government communication can empirically alter household energy behavior remains an understudied question. While a small body of work examines communication effects during natural experiments (Reiss and White, 2008; Ruhnau et al., 2023; Jamissen et al., 2024; Behr et al., 2025) systematic evidence tracing the full causal chain from government messaging to public attention to measurable demand response is scarce (He and Tanaka, 2023).

This paper fills the gap by focusing on France and exploiting the natural experiment provided by the 2022-23 European energy crisis. Triggered by a combination of geopolitical tensions, including the curtailment of Russian gas imports, and technical constraints, such as the temporary shutdown of nuclear reactors, the crisis put the French electricity system under unprecedented stress. Wholesale prices surged, public concern over potential winter power failures intensified and the focus of the nation shifted towards electricity consumption. Household demand was one of the few areas where rapid adjustment could be made to maintain system balance. In response to this situation, the French government introduced two non-monetary policies aimed at reducing electricity consumption. Firstly, they launched a nationwide communication campaign urging citizens to reduce their energy use. This campaign included televised appeals and consistent messaging promoting low-cost conservation measures, such as lowering indoor temperatures to 19°C. Secondly, the Minister for Energy Transition authorised the national grid operator to implement remote load control on households with specific electric heating systems. This intervention temporarily reduced the electricity consumption of these households during peak demand periods, thereby limiting their capacity for voluntary adjustment.

The aim of this paper is to quantify the extent to which government communication can alter household electricity demand by tracing the mechanism linking communication to consumption, and examining its impact on public attention.

The identification strategy is based on the concept that wholesale electricity prices affect household demand via two distinct channels in France. The first channel is institutional and operates through the national regulated retail tariff, which adjusts with substantial lags of typically six months or more. This channel may also be partially decoupled from wholesale prices through ministerial interventions, such as the tariff shield implemented during the 2022-23 period. The second, faster channel is hypothesised to operate through government communication: sharp increases in spot prices are followed by crisis and conservation oriented messages that shape public attention and potentially household behaviour within weeks. While announcements about tariff setting may generate anticipation effects, these are likely to be mediated by the same informational environment, as expectations about future price changes are primarily conveyed through public communication. Consequently, anticipation is subsumed within the attention channel rather than constituting a separate pathway. This channel is empirically identified using high-frequency variation in attention, which captures rapid fluctuations in information exposure and precedes effective changes in retail prices. This allows the estimation to isolate a fast-moving behavioural response alongside the slower institutional pass-through.

A three-step empirical strategy was employed to identify the chain from government communication to household electricity demand. First, over 12,000 official statements were collected from the government's central website, which logs all speeches and media appearances by government members. A semi-supervised machine learning algorithm was then employed to categorise these communications by narrative type, yielding daily measures of communication intensity. Additionally, the INA Lab helped retrieve the daily number of public service announcements (PSA) from the televised campaign launch during winter 2022-23.[1] Secondly, the link between these narratives and public attention is established by combining the communication series with Google Search Volume (GSV) and using a reduced-form model to distinguish between conservation and crisis signals. Thirdly, the impact of narrative-specific attention on household electricity usage is estimated using a time-series framework that makes use of the daily frequency of the data. The high-frequency setting is

[1] The Institut national de l'audiovisuel (INA) is France's national audiovisual archive and research institution.

crucial here: while policy interventions occurred simultaneously during the crisis, communication shocks materialised within days. In contrast, retail price adjustments evolved more slowly. This sequential design traces the mechanism from communication to attention and ultimately to consumption.

The empirical results highlight three key findings. Firstly, the effectiveness of communication depends fundamentally on framing. Conservation messaging only gained traction when embedded within broader crisis narratives that emphasised the economic consequences. Conversely, attention to crisis-related communication is a direct response to concerns about system reliability and peaks during periods of intense media coverage of potential shortages. Secondly, attention only translates into a demand response when households possess actionable flexibility. Crisis-related attention produced uniform but modest reductions in demand across all household types. By contrast, conservation-related attention exhibited stark heterogeneity depending on the level of flexibility available to households. Thirdly, prices and temperatures quantitatively dominated demand responses. Although communication contributed measurably during moments of acute crisis, its impact was modest compared to the persistent influence of economic incentives and climatic conditions. Taken together, these findings suggest that, although communication can mobilise attention, its translation into behavioural change critically depends on narrative framing and technological constraints that shape demand-side flexibility.

This paper makes two key contributions. Firstly, it provides rare causal evidence of the effectiveness of government communication in shaping household electricity demand during a large-scale energy crisis. The analysis shows that communication-driven attention reduced residential electricity demand by between 1 and 2%. This estimate is lower than that of the meta-analysis of randomised studies by Buckley (2020), who found that non-monetary conservation incentives reduce energy consumption by 2–4%, and lower than the estimate of an 8% decrease in residential energy consumption by Behr et al. (2025). Second, the paper estimates a residential electricity price elasticity of 0.2 during crisis conditions, which is substantially lower than the elasticities typically reported (Auray et al., 2019; Csereklyei, 2020). This finding contributes to an emerging body of literature which demonstrates that failing to account for non-monetary motivations in estimation strategies can lead to the overestimation of price elasticity by attributing non-price driven savings incorrectly to price induced responses (Ito et al., 2018; Ruhnau et al., 2023; Jamissen et al., 2024).

The results have broader implications for the design of demand-side policy. Informational campaigns are most effective when they coincide with systemic shocks that heighten public concern, and they have the greatest impact on consumer segments with sufficient discretionary flexibility. While communication serves as a low-cost, rapidly deployable crisis tool, it only amplifies existing adjustment capacity rather than creating it. Long-term decarbonisation strategies must integrate communication with systematic efforts to democratise access to demand-side flexibility. This will ensure that future appeals engage the broader residential sector, rather than only targeting the minority who are already flexible.

Taking a step back, while the crisis context strengthens identification, it also narrows the scope of external validity. The underlying mechanisms are likely to extend beyond France, though their magnitude and expression depend on context. Periods of heightened salience, characterised by immediate risks and strong media attention, appear particularly effective at capturing public attention. This suggests that informational campaigns can be especially powerful when they resonate with perceived urgency, and that adapting communication strategies could improve their effectiveness even for more gradual challenges such as climate change. Institutional and technological features further shape these dynamics. In France, the relatively slow pass-through of prices create conditions where communication can play a prominent role alongside price signals. In other settings, such as systems with real-time pricing, behavioural responses may be more directly influenced by prices.

The remainder of this paper is organised as follows: Section 2 presents energy conservation definition. Section 3 presents the background of the residential electricity market in France. Then, Section 4 presents the sources, pre-processing and description of datasets used for this empirical study. Section 5 presents the empirical methodology and Section 6 the main results.

2 Definitions and Literature around Energy Conservation at Home

The latest IPCC report defines the concept of sufficiency as *policies, measures, and daily practices that avoid the demand for energy, materials, water, and land while delivering human well-being for all within planetary boundaries* (Shukla et al., 2022). In the residential sector, sufficiency regarding at home energy use is usually referred to as energy conservation. It seeks to reduce final energy use through changes in habits, behaviors, and consumption patterns, rather than through technological improvements alone.[2] This paper adopts the language of conservation, as the empirical setup does not allow for assessing broader implications for planetary boundaries.[3]

[2](See for exp. Richler, 2016; Jachimowicz et al., 2018; Myers and Souza, 2020; Bonan et al., 2021; Knittel and Stolper, 2021; Caballero and Ploner, 2022; Loschke et al., 2024)

[3]Energy efficiency refers to technological improvements that reduce energy consumption per unit of service, though constrained by the energy paradox (Jaffe and Stavins, 1994) and rebound effects (Sorrell et al.,

In a formal economic framework, List et al. (2023) conceptualize behavioural biases as internalities, systematic errors in how agents perceive the marginal benefits or costs of consumption. These internalities often stem from overlooked co-benefits, poor information, or limited awareness of the broader consequences of consumption choices. In this context, energy conservation incentives can be interpreted as mechanisms that reduce these internalities, by providing information, setting goals, or reshaping social norms. This theoretical perspective aligns with earlier research suggesting that individuals often lack sufficient information to engage in optimal energy-saving behaviors, and that acquiring such information can be costly (**?**De Young, 2000; Hungerford and Volk, 1990; Schultz et al., 2002). behavioural incentives are therefore particularly potent during crises, as crisis framing acts as a salience amplifier in an information-rich environment where attention is scarce (Kudesia and Lang, 2024). When messages emphasise imminent risks, such as power shortages or blackout probabilities, they cut through competing stimuli, elevate issue salience, and trigger rapid information seeking and behavioural vigilance (Curotto et al., 2025; Spence et al., 2021). This is consistent with classic attention-economics logic and media-effects research on agenda-setting (Dyer and Kolic, 2020; Scheufele and Tewksbury, 2007; Wicke and Bolognesi, 2020).

While theoretical arguments support the effectiveness of conservation incentives, the empirical evidence reveals considerable heterogeneity in their actual impact. Several meta-analyses have attempted to classify conservation incentives and explain variations in their effectiveness (Abrahamse et al., 2005; Andor and Fels, 2018; Blasco and Gangl, 2023; Delmas et al., 2013; Brandon et al., 2017). Building on these reviews, four broad categories of incentives emerge: Monetary incentives, which aim to align consumption behaviors with financial self-interest by making costs and potential savings more salient. These incentives tend to be more effective when the potential gains are significant (Delmas et al., 2013). Goal-setting interventions, which involve setting specific consumption targets (e.g., "reduce your consumption by 10%"). While goal-setting alone shows limited effectiveness, its impact increases substantially when combined with feedback mechanisms (Abrahamse et al., 2005; Andor and Fels, 2018). Feedback incentives, which provide information about energy use. Feedback can relate to past consumption, real-time consumption, or comparative consumption relative to peers. Comparative feedback, leveraging social norms, is found to be the most effective (Delmas et al., 2013). Information strategies, which disseminate energy-saving tips. Low-involvement strategies (e.g., mass media campaigns, general workshops) have limited behavioural impact, while high-involvement strategies (e.g., personalised advice after a home

2007; Peñasco and Anadón, 2023). Energy conservation involves behavioural adjustments to deliberately reduce consumption (Gardner and Stern, 1996), making it less exposed to rebound effects and critical for demand-side management.

energy audit) are more successful (Gonzales et al., 1988; Winett et al., 1982). Quantitatively, Buckley (2020) estimates that earlier reviews suggested energy savings of around 7% could be achieved through informational incentives. However, using a stricter selection of more recent experimental studies, she argues that a more realistic expectation is a reduction of 2–4% in energy consumption. Beyond the classification of incentives, another important issue is the persistence of effects. Allcott and Rogers (2014) emphasise that the effects of behavioural interventions often decay over time unless reinforced. Their work highlights the importance of repeated interventions initially, to help individuals form new habits, before reducing intervention intensity once behaviors are stabilised.

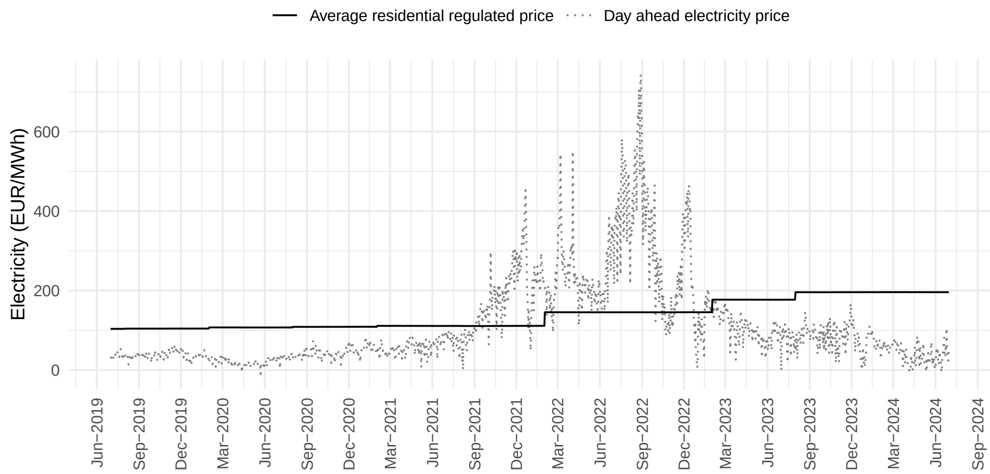
Although conservation messaging felt novel during the 2022 energy crisis, incentives for energy saving have long existed. On 2 February 1977, during the oil crisis, President Jimmy Carter gave a televised speech urging Americans to lower their thermostats to 65°F (18°C) during the day and 55°F (13°C) at night to save natural gas. Following this announcement, Luyben (1982) assessed the effectiveness of the televised appeal and found no significant difference in thermostat settings between those who had heard the message and those who had not. More broadly, several studies in the late 1970s and 1980s investigated the effectiveness of conservation incentives (Craig and McCann, 1978; Hutton and McNeill, 1981; Walker, 1980). In France, conservation campaigns have also been launched repeatedly since the first oil shock of 1973. Initially motivated by energy security concerns, the Ministry of Energy and the newly created *Agence pour les Économies d'Énergie* (AEE) promoted reduced heating and efficiency measures. The most emblematic slogans were "In France we don't have oil, but we have ideas" (*En France on n'a pas de pétrole mais on a des idées*, 1976) and the "Hunt for waste" (*Chasse au gaspi*, 1980s). With the rise of climate concerns and the Kyoto Protocol, conservation returned to the policy agenda in the late 1990s. The *Agence de l'environnement et de la maîtrise de l'énergie* (ADEME), created in 1990, launched new national campaigns, most notably "Energy savings, let's move fast, it's getting hotter !" (*Économies d'énergie, faisons vite, ça chauffe !*, 2004–2006), which framed conservation as both an environmental necessity and a means of reducing household costs. Unlike these earlier efforts, typically mediated by specialised agencies such as ADEME, the 2022-23 campaign was spearheaded directly by the government at the highest political level. This shift reflected both the severity of the crisis and the need for rapid, visible coordination, making it one of the largest conservation appeals ever conducted in Europe.

3 Background

3.1 Residential Electricity Market in France

Since July 2007, the French electricity retail market has been fully opened to competition for all consumers, including residential customers. As a result, clients now have access to two main types of contracts: regulated tariff contracts (*Tarifs Réglementés de Vente*, TRV) and market-based contracts (*Offres de Marché*, OM). The TRV is defined by the energy regulatory commission (*Comission de Régulation de l'Energie*, CRE) and is offered by the historical supplier (*Électricité de France*, EdF). Market offers, on the other hand, are proposed by both historical and alternative suppliers. According to the CRE's annual evaluation report, by the end of 2023, 62% of residential sites were under TRV contracts, while 14% had a market-based contract linked to a TRV. As a result, 76% of households were dependent on regulated tariffs (CRE, 2024).

The concept of regulated tariffs was introduced in France in 1946 to ensure universal access to electricity, with a key objective of maintaining price stability for consumers. These tariffs are governed by the *Code de l'énergie* (Article, 2011, 2015), which outlines the regulatory framework for their implementation and adjustment. Since 2016, the responsibility for proposing TRVs has shifted to the CRE, which submits a revised set of tariffs to the French Ministers of the Economy and of Energy every six months. The ministers then have three months to accept or reject these proposals. In the absence of any response within this period, the proposed tariffs are considered implicitly accepted. Figure 1 presents day-ahead electricity prices and average residential TRVs, illustrating that the regulatory mechanism allows for decoupling between wholesale prices and retail tariffs.



Notes : This figure displays daily day-ahead electricity prices (gray line) and average monthly regulated residential tariffs (TRV, black line) from June 2019 to September 2024. Day-ahead prices represent wholesale spot market transactions. Regulated residential tariffs reflect the average price paid by households, net of taxes and network charges, incorporating only the energy component.

Figure 1: France Electricity Market and Regulated Prices

3.2 The 2022-23 Energy Crisis: Context and Implications for France

The European energy crisis of 2022-23 exposed the vulnerability of energy supply systems to geopolitical shocks. In France, the combination of reduced Russian gas supplies and declining nuclear generation capacity put acute pressure on the electricity market.[4] As documented by the CRE, these tensions led to unprecedented increases in regulated electricity tariffs, threatening household purchasing power despite traditionally stable prices in France.

In response to the energy crisis, the French government adopted two distinct yet complementary types of policy instrument. First, a monetary instrument: the government implemented a price shield mechanism to mitigate the impact of rising wholesale prices on household electricity bills. At the end of 2021, when faced with an average proposed increase in regulated tariffs of 44.5%, the government acted swiftly. The domestic tax on final electricity consumption (TICFE) was reduced drastically to €1/MWh for all consumers. Average retail price increases were capped at 4% including taxes in February 2022. Secondly, the government introduced non-monetary instruments. Although the first political signals

[4] At the time, France's nuclear fleet was severely constrained by extended maintenance and stress corrosion issues, resulting in nuclear output reaching its lowest level in decades. As of late 2022, nearly half of the approximately 56 reactors were offline, significantly reducing annual production, according to the annual report from the French Energy Regulatory Commission.

emerged in February 2022, when President Emmanuel Macron declared energy conservation to be a national priority, the energy conservation plan was officially announced on 6 October 2022. In June 2022, the government set an initial target of reducing energy consumption by 10% by 2024, initially focusing on the public sector, industry, and services, before extending this to all businesses and households in September 2022. This was accompanied by an official launch and a national communications campaign entitled 'Every action counts', which was broadcast on television, radio and social media. The campaign aimed to encourage households to adopt energy-saving behaviours and promote practical, everyday best practices. In addition to broadcasting these adverts, government members frequently referenced energy conservation and the energy crisis in their daily public statements. In addition, the plan included a technical measure targeting residential electricity consumption specifically. For around 4 million households with programmable hot water tanks and a Time-of-Use tariff contract, accounting for 28% of residential contracts between 2019 and 2024, network operators were authorised to delay water heater activation by up to two hours between 11:00 and 15:30, provided the delay began before 14:00. This measure was implemented over two consecutive winter seasons: from October 2022 to April 2023 and from November 2023 to March 2024.

Although the monetary instrument limited the full pass-through of market price increases, regulated tariffs still rose during this period, thus preserving at least some incentives for reducing demand. This combination of policies protects households from the sharpest price shocks while simultaneously encouraging behavioural adjustments. It reflects a deliberate strategy to balance affordability concerns with energy security objectives. Rather than being contradictory, jointly deploying monetary and non-monetary levers illustrates the government's attempt to manage crisis-induced demand pressures without undermining public support.

4 Data

Data on residential electricity consumption are obtained from Enedis (2024). Enedis, a regulated subsidiary of EDF, manages the low- and medium-voltage grid covering roughly 95% of households in mainland France, with the remainder served by local municipal distributors. As a complement, a small household panel data is used, Enertech (2022), allowing to link electricty consumption profils with socio-economic insights. Temperature is drawn from ECMWF (2024). Price data combine two sources: regulated retail tariffs provided by Commission de Régulation de l'Energie (2024) and day-ahead wholesale electricity prices retrieved from ENTSO-E (2024). To measure government communication, the study compiles more than 12,000 official statements from Vie-publique.fr (2024) public repository, which archives all public speeches by French government members. As a complement, public interest advertising (PSA) are retrieve from the French audiovisual archives (INA) (Institut national de l'audiovisuel, 2024). These are combined with daily Google Search Volume (GSV) indices for energy-related terms, which proxy public attention over time (Google Trends, 2024). Table 11 in Appendix C summarises all data sources, frequency, and coverage.

4.1 Residential Electricity Consumption

This study uses daily electricity consumption data from Enedis, the main low-voltage distributor in France, as the main source. The data cover the period from July 2019 to June 2024. The data represent national aggregate consumption and total contracts for two representative household types: those on a base tariff and those on a Time-of-Use (ToU) tariff. These different tariff structures are intended to offer consumers options that reflect their usage patterns, while encouraging electricity consumption during ToU hours to help balance grid demand (Enedis, 2024). The years are defined as July-to-June energy years in order to capture the entire winter heating season within a single observation period. Additionally, the Elecdom panel is used to characterise consumer segments by tariff and contracted power. This panel comprises a representative sample of French households with connected measurement systems that record electricity consumption at 10-minute intervals. Each household is monitored across an average of 24.8 measurement points, covering all major end-use categories (Enertech, 2022).

Table 1, Panel A. illustrates the annual electricity consumption patterns for these two profiles. Households with ToU contracts, who benefit from lower period rates, exhibit significantly higher electricity consumption compared to base profile households, whose average annual consumption is only about half that of ToU users. This difference in consumption level is explained by end-use type for the different tariff profiles. Table 1, Panel B. showcases that for ToU the cumulative use of home and water heating represent 72% of consumption i.e 4.2 MWh/year, while for base it represents 60% of consumption i.e 1.7 MWh/year.

A marked reduction in electricity consumption is observed when comparing the pre-crisis reference period (2019-21) with the crisis years (2022-23). Overall, residential consumption decreased by 13.8 TWh. This reduction was entirely driven by households with ToU contracts, who reduced their consumption by 12.7 TWh. In contrast, base profile consumption decreased by just 1.1 TWh over the same period. However, when the data is expressed in terms of consumption per contract, a more nuanced picture emerges. After adjusting for the 2% increase in the number of base contracts between 2021 and 2023, base profile households demonstrate a greater decline in consumption. Specifically, the reduction amounts to 0.9 MWh/contract/year for ToU households and 0.1 MWh/contract/year for base households.

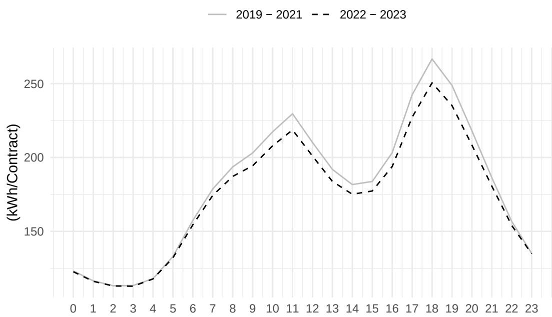
Table 1: Average Annual Electricity Consumption

	Base		ToU	
	2019–2021	2022–2023	2019–2021	2022–2023
Panel A. Aggregate Consumption				
Total consumption (TWh)	51.0	49.9	97.7	85.0
Number of contracts (millions)	17.6	18.0	14.5	14.6
Consumption per contract (MWh)	2.9	2.8	6.7	5.8
Panel B. End-Use Composition (%)				
Heating	25		48	
Hot water	35		24	
Refrigeration	21		10	
Cooking	10		6	
Other appliances	9		12	

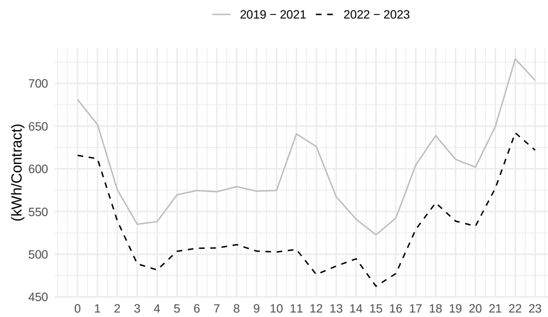
Notes : Years are defined from July to July in order to encompass one winter period per year of study. Thus, the year 2022 goes from July 2022 to June 2023.

To go further on the description of residential electricity during the period, Figure 2 presents the average hourly electricity consumption per contract during winter months for households on and ToU tariffs. The goal is to visualise the effect of load control, one of the two non-monetary measures implemented during the crisis. ToU households display a marked drop in midday consumption during the intervention period. Between 11:00 and 14:00, the load curve for 2022-2023 shows a distinct downward shift relative to the pre-intervention baseline, consistent with the remote deactivation of water-heating systems during this period. The reduction is sharpest between 12:00 and 14:00, corresponding closely to the maximum two-hour control window authorised under the policy.

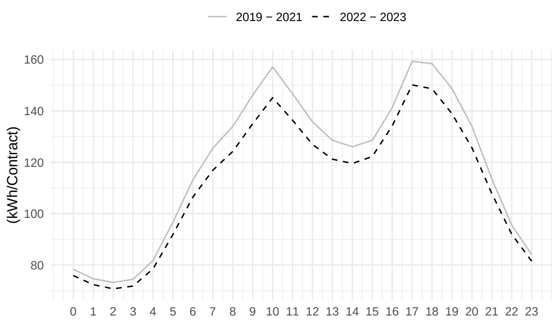
This visual evidence strongly suggests that the load-control measure explains the reduced consumption during the targeted hours without significantly shifting demand to adjacent hours. The difference between base and ToU profiles reinforces the interpretation that the observed midday reduction is directly attributable to the hot-water control policy, since base tariff households were not subject to the intervention. For comparison, the same daily load profiles are shown for the summer months. As expected, the patterns lack the midday dip observed in winter, reinforcing once more the idea that the reduction between 11:00 and 14:00 during winter is specifically associated with the controlled deactivation of hot water systems rather than an individual shift in consumption.



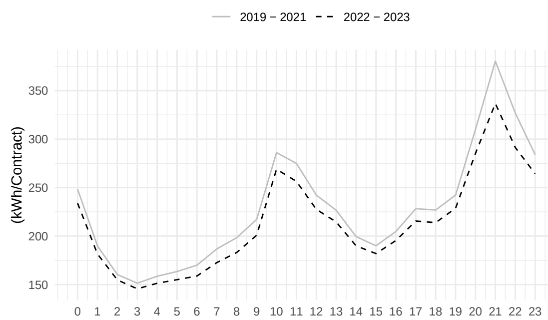
(a) Winter - base



(b) Winter - ToU



(c) Summer - base



(d) Summer - ToU

Figure 2: Average hourly load by season and tariff profile

4.2 Retail Tariff

Daily data on retail electricity prices for residential consumers are not available in France. However, given that approximately 76% of residential customers are, directly or indirectly, subject to the regulated electricity tariff, this study relies on available bi-annual TRV data to approximate the evolution of retail prices.

The residential sector is characterised by different tariff profiles, primarily distinguished by the contractual options for electricity pricing. Table 2 Panel A presents the average daily distribution of contracts for the period 2019-2024, across tariff profiles and subscribed power levels. The base tariff profile accounts for 54% of residential contracts, while the ToU profile represents 45%. The most common configuration is the base profile with a 6 kVA subscription, followed by the ToU profile with the same subscribed power. Together, these two tariff and power combinations represent around 59% of all contracts in the database. In contrast, the tempo profile, characterised by variable pricing based on daily color-coded signals, accounts for only 1% of contracts. Table 2 Panel B presents the average daily distribution in terms of electricity consumption for the period 2019-2024, across tariff profiles and subscribed power levels. The base tariff profile accounts this time for 34% of residential consumption, while the ToU profile represents 64%. The strongest consumption comes from ToU profile with a 9 kVA subscription, followed by the base profile with a 6 kVA subscription. Together, these two tariff and power combinations represent around 39% of total consumption. Once again the tempo profile, accounts for less than 2% of consumption. Due to its complex pricing structure and limited representation in the data, tempo contracts are excluded from the analysis.

Table 2: Distribution of Residential Electricity Contracts and Consumption

Panel A. Share of Residential Contracts (%)										
	Contracted Power (kVA)									
	3	6	9	12	15	18	24	30	36	Total
Base	4.46	**39.73**	7.33	1.96	0.30	0.37	0.05	0.02	0.06	**54.28**
ToU	–	**19.16**	15.81	6.70	1.00	1.66	0.16	0.05	0.12	44.66
Tempo	–	–	**0.70**	0.16	0.03	0.15	–	0.01	0.01	1.06
Total	4.46	**58.89**	23.84	8.82	1.33	2.18	0.21	0.08	0.19	100.00
Panel B. Share of Residential Electricity Consumption (%)										
	Contracted Power (kVA)									
	3	6	9	12	15	18	24	30	36	Total
Base	1.08	**21.24**	7.07	3.06	0.54	0.97	0.19	0.08	0.33	34.56
ToU	–	17.60	**24.94**	13.45	2.28	3.97	0.67	0.24	0.64	**63.79**
Tempo	–	–	**0.86**	0.31	0.06	0.32	–	0.06	0.04	1.65
Total	1.08	**38.84**	32.87	16.82	2.88	5.26	0.86	0.38	1.01	100.00

Notes : Panel A reports the share of residential electricity *contracts* by tariff type and contracted power (kVA). Panel B reports the corresponding share of total *electricity consumption*. Bold values indicate the modal power level within each tariff profile. Dashes (–) denote non-applicable combinations of tariff and contracted power.

To build consistent price series for empirical analysis, two weighted tariffs are constructed: one for households with a base profile and another for households with a ToU profile. The underlying data for this construction are the set of 18 TRV tariffs provided by the CRE, corresponding to different combinations of profile and subscribed power levels. The weighting is based on the observed distribution of contracts in the Enedis database (see Appendix C Table 12 for the yearly rate change of contracts by profile). The weighted tariff at day t, denoted $P^w_{i,t}$, is calculated according to the following formula:

$$P^w_{i,t} = \frac{\sum_{i=1}^{n} P_{it} \times w_{it}}{\sum_{i=1}^{n} w_{it}} \tag{1}$$

where P_{it} is the tariff for profile and power i at time t, and w_{it} is the weight reflecting the proportion of consumption corresponding to that tariff. This methodology enables the construction of two bi-annual weighted tariffs that accurately reflect the market structure for base and ToU customers.

Table 3 presents the evolution of residential electricity tariffs for both base and ToU profiles across the crisis period. The tariff structure can be decomposed into four distinct pricing blocks. The upstream period represents the pre-crisis baseline, characterised by relatively stable prices averaging 98.06 €/MWh for the variable component and approximately 9–12 €/month for fixed subscription fees. Following the onset of the energy crisis in mid-2022, tariffs increased sharply across three successive adjustment periods. By the third period, variable tariffs had risen by 92% for base households and 81% for ToU households relative to the upstream baseline. Notably, the government's tariff shield substantially attenuated these increases: absent the intervention, the CRE recommended tariffs would have imposed increases of 163% (base) and 152% (ToU) for variable charges alone. The divergence between observed and recommended tariffs was most pronounced during the second period, when recommended variable prices reached 314.00 €/MWh, more than triple the upstream level, while observed prices rose to only 170.80 €/MWh for base households, reflecting the tariff shield's maximum protective effect. This institutional cushioning limited household exposure to wholesale market volatility while still transmitting a substantial price signal.

Table 3: Tariffs for base and ToU profiles

	Observed Prices		Recommended Prices (CRE)		
	Variable (€/MWh)	Total (€/MWh)	Variable (€/MWh)	Total (€/MWh)	Fixed (€/Month)
Panel A. Base					
Upstream	98.06	107.12	98.06	107.12	9.05
First period	137.40	147.58	153.41	163.81	10.39
	(+40.1%)	(+37.8%)	(+56.5%)	(+52.9%)	(+14.8%)
Second period	170.80	181.60	314.00	324.80	10.80
	(+74.2%)	(+69.5%)	(+220.2%)	(+203.2%)	(+19.3%)
Third period	188.70	200.11	258.13	269.54	11.41
	(+92.4%)	(+86.8%)	(+163.2%)	(+151.6%)	(+26.1%)
Panel B. Peak/Off-Peak					
Upstream	98.14	109.80	98.14	109.80	11.65
First period	130.35	145.92	159.26	13.35	
	(+32.8%)	(+30.7%)	(+48.7%)	(+45.1%)	(+14.6%)
Second period	159.15	172.79	302.35	315.99	13.64
	(+62.2%)	(+57.4%)	(+208.1%)	(+187.8%)	(+17.1%)
Third period	177.65	191.91	247.08	261.34	14.26
	(+81.0%)	(+74.8%)	(+151.7%)	(+138.0%)	(+22.4%)

Notes : Variable prices represent the per-MWh energy charge; fixed prices represent monthly subscription fees. Observed prices reflect actual tariffs applied to consumers, incorporating the government's tariff shield. Regulated prices (CRE) show the Energy Regulatory Commission's recommended tariffs absent intervention. Percentages in parentheses indicate changes relative to the upstream (pre-crisis) period. Upstream period: pre-August 2022; First period: August 2022–January 2023; Second period: February–July 2023; Third period: August 2023 onward.

4.3 Weather Components

Given the strong thermosensitivity of residential electricity consumption it is essential to control for weather conditions when analysing consumption dynamics. In particular, the significant seasonal fluctuations in consumption are largely driven by outdoor temperatures, with consumption typically around 20% higher during the winter months compared to the summer months.

To capture the main meteorological drivers of electricity demand, the outside temperature is measured as the 2-meter temperature above the surface, in degrees Celsius. Temperature is the dominant factor in explaining electricity demand fluctuations in France, due to the high prevalence of electric heating, which covers approximately 32% of the housing stock

(Bouton, 2024). This relationship between temperature and energy use is highly asymmetric: while colder weather strongly increases consumption, warmer summer temperatures do not substantially reduce it (Henley and Peirson, 1997). As a result, following standard practice, Heating Degree Days (HDD) are constructed rather than using raw temperatures. The HDD at a given day t are defined based on a reference base temperature of 15°C, the official threshold used for France.[5] Formally, HDD are calculated according to:

$$HDD_t = \begin{cases} 15 - T_t & \text{if } T_t < 15 \\ 0 & \text{otherwise} \end{cases} \tag{2}$$

Thus, HDD capture the number of degrees below the heating threshold, reflecting the likely heating demand. Cooling Degree Days (CDD) are not considered in this study, as there is no evidence of significant cooling behaviours at the national level for the residential sector in France, see Allcott and Rogers (2014) for an estimation of a cooling effect. Spatial aggregation of weather data is performed using a population-weighted approach, consistent with best practices in the literature (see for example Kennard et al., 2022). Each weather observation from the spatial grid is weighted by the associated population, using the most recent census data available, according to:

$$weather_s(pop) = \frac{weather_s \times pop_s}{\sum_{i=1}^{s} pop_i} \tag{3}$$

where $weather_s$ is the meteorological variable for cell s, pop_s is the population in that cell, and $\sum_{i=1}^{s} pop_i$ is the total population over all cells. This method ensures that national-level weather indicators reflect the conditions experienced by the majority of the population, thereby better capturing the effective drivers of aggregate residential electricity consumption.

An important pre-processing element to consider is that the daily electricity consumption data used in this study are seasonally pre-adjusted, notably for standard seasonal patterns such as weekly and holiday effects. However, they are not corrected for daily weather anomalies relative to historical norms. To isolate the effect of unusual temperature conditions, the daily HDD series is therefore centred on its historical average. This approach allows for measuring deviations in heating needs for a given day, relative to normal conditions. In Appendix B Figure 9 showcases, among other variables, the two pre-adjusted temperature

[5] The choice of 15°C as the base temperature for Heating Degree Days (HDD) in France is aligned with the national standards used by energy authorities and network operators. It reflects the temperature below which households typically start heating their dwellings. While some international studies use different thresholds, ranging from 15°C to 18°C depending on the country's building stock and heating practices (Staffell et al., 2023), the 15°C benchmark is considered appropriate for the French residential sector, given previous calibration of heating models for France (Bruguet et al., 2025).

and electricity consumption time-series.

4.4 Government Communication and Public Attention

To trace the mechanism from government messaging to household behavior, two sets of measures are constructed: official communication intensity and public attention to energy narratives.

Communication. Government communication is captured using all official statements published on the French government's central platform between January 2019 and December 2024. This repository archives over 160,000 speeches and communications from the President, Prime Minister, and government members since the 1970s. After systematic scraping and cleaning, the corpus comprises 12,184 documents, each containing the full text, speaker identity, position, and delivery date (Vie-publique.fr, 2024). A semi-supervised machine learning algorithm is applied to classify each statement into three narrative categories. *Conservation* communication promotes voluntary energy demand reductions and behavioral adjustments. *Crisis* communication emphasises price volatility, and potential disruptions. *Both* captures statements combining conservation appeals with crisis framing, with the idea that narratives around scarcity can trigger behavioural change (Curotto et al., 2025; Spence et al., 2021; Jamissen et al., 2024; He and Tanaka, 2023). To systematically analyse government communication in promoting residential energy savings during the 2022-23 crisis, it is essential to quantify the flow and intensity of relevant public communication over time. Given the scale of the corpus and the diversity of political discourse, manual labeling alone would be impractical. Therefore, a classification strategy is developed to automatically identify and track government communication encouraging energy savings.[6] To complement these official statements, televised campaign intensity is measured through public service announcements retrieved from France's national audiovisual archive, the Institut National de l'Audiovisuel (INA) (Institut national de l'audiovisuel, 2024). Public interest advertising (PSA) refers to government-sponsored communication campaigns designed to promote socially desirable behaviours rather than commercial outcomes. In France, such campaigns are coordinated by public agencies or ministries and disseminated through major media channels at no or reduced cost. Unlike commercial advertising, PSAs pursue non-market objectives such as improving health, safety, or energy conservation. Legally, they are classified as non-commercial messages under the French broadcasting framework and may be aired outside ordinary advertising quotas, provided they respect general content restrictions on decency, truthfulness,

[6]The full methodology regarding communication time-series production is detailed in Appendix A.

and political neutrality. Economically, these campaigns serve as a low-cost policy lever to influence individual behaviour when direct regulation, taxation, or price signals are insufficient or politically costly. Their effectiveness depends on attention formation, salience, and social norm mechanisms rather than pecuniary incentives, positioning them at the intersection of behavioural economics and public policy.[7]

To characterise the government communications to which households were likely to react, a structured topic modeling approach was applied to the corpus. This method identified the dominant themes within these communications. In the case of energy conservation, government communications were structured around themes related to ecological transition (ecological transition, biodiversity), housing renovation (housing, renovation, communities), and decarbonised energy technologies (hydrogen, nuclear, renewables). These groupings reflect an official tone oriented toward long-term planning, particularly the implementation of sustainability policies concerning energy policy. Two sub-themes were also identified on a cyclical basis during the winter of 2022-23: the Russo-Ukrainian war (armies, european, ukraine) and the role of the civil service (civil service, civil servants), highlighted as an exemplary actor in conservation efforts. Communications relating to the energy crisis are distinguished by a more pronounced emphasis on economic and geopolitical urgency. Government communication focused primarily on the cost of the crisis, particularly for local authorities (euros, municipalities, communities, millions). The term euros appears more than a thousand times, thus underscoring the centrality of the price argument. However, from December 2022 onward, crisis communication shifted decisively toward warnings of imminent power failures, as evidenced by nine governmental statements (December 5–13) explicitly addressing blackout risks.[8]

Attention. Public attention to energy narratives is proxied using Google Search Volume (GSV) indices (Google Trends, 2024). GSV measures the relative search intensity of query terms on a normalized 0–100 scale, where 100 represents peak search interest during the queried period. Indices are then constructed using sets of related search queries rather than single keywords, to mitigates measurement noise from idiosyncratic spikes in single search terms and yields a broader, more stable proxy of digitally expressed public attention (see Appendix C.2). An important caveat is that GSV captures search behavior only among internet users. In the French context, this likely under-represents older households and rural populations with lower internet access rates, while over-representing younger, urban, and

[7]Law no. 86-1067 of 30 September 1986, relating to freedom of communication, which establishes the general framework for audiovisual communication in France.

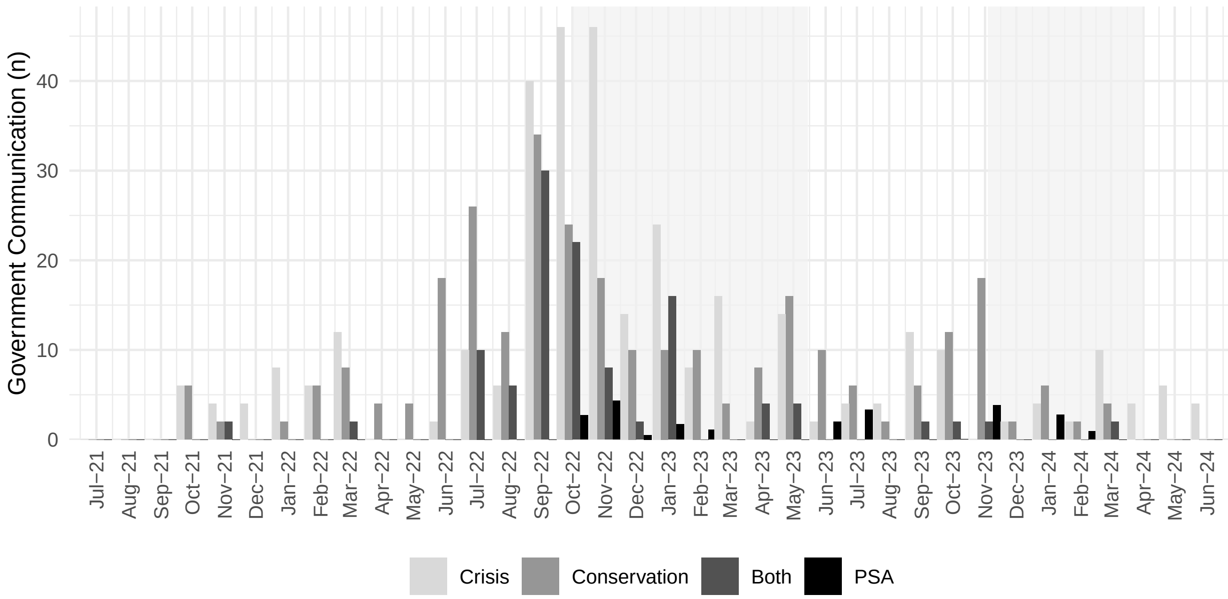
[8]See for example an interview with the Minister for Ecological Transition emphasising power failure risks and urging households to prepare with batteries and generators.

digitally connected demographics. The measure should therefore be interpreted as reflecting digitally expressed attention rather than universal public awareness or comprehension of energy issues.

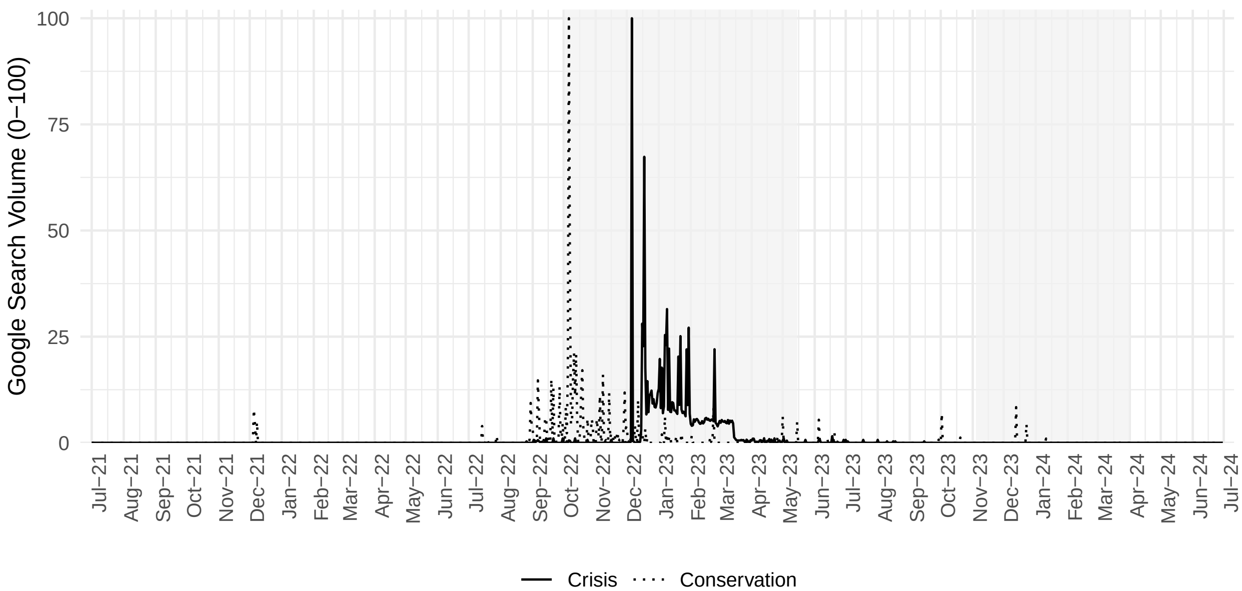
Descriptive insights. Figure 3 shows how government communication and public attention evolved during the crisis period. The shaded areas indicate periods of load control, during which grid operators managed residential water heating systems remotely.

Panel A shows the monthly number of official communications categorised by narrative type. Both the conservation and crisis narratives exhibit a normal distribution; however, crisis communication dominated in late 2022, which coincided with peak energy price volatility and supply concerns. Television PSAs (scaled by 100 for visibility) were deployed intensively during two consecutive winter periods, with additional campaigns in the summer of 2023 (July–August) emphasising the conservation of cooling devices.

Panel B displays daily Google Search Volume indices. Attention to conservation shows gradual variation with sustained elevation during the first load control period. Crisis attention exhibits greater volatility and strong reactivity to policy announcements. Notably, crisis-related searches surged sharply from 6 December 2022, immediately following the nine official statements warning of imminent risks of household blackouts. This suggests that explicit crisis framing may trigger more immediate public engagement than conservation appeals alone.



(a) Government Communication



(b) Public Attention

Notes : Panel A shows the monthly count of government communications by narrative type from July 2021 to June 2024. *Crisis* includes communications emphasising supply risks and price volatility. *Conservation* covers messages promoting voluntary energy reductions. *Both* combines crisis and conservation themes. *PSA* represents public service announcements aired on television (scaled to n/100 for visualisation). Shaded areas indicate load control periods. Panel B displays daily Google Search Volume indices for crisis-related and conservation-related queries.

Figure 3: Communication Campaigns and Attention Formation

5 Empirical Strategy

5.1 General Identification Strategy

The identification strategy exploits the institutional structure of the French electricity market, where wholesale price movements affect residential consumption through two distinct channels operating at different frequencies.

The price channel. Regulated retail tariffs adjust slowly and imperfectly to wholesale market conditions. Under normal circumstances, revisions occur with a lag of at least six months relative to spot price movements, since the energy regulator must evaluate pass-through proposals before they can be approved by ministers. During the 2022-23 crisis, this institutional friction was exacerbated by the tariff shield policy, which capped retail price increases despite sustained wholesale volatility. Consequently, the price channel is slow-moving, mediated by administrative decisions and, at times, partially disconnected from underlying market dynamics.

The attention channel. Wholesale price spikes triggered an immediate government response in the form of intensive communication campaigns emphasising the severity of the crisis and the urgency of conservation, reinforced by coordinated public service announcements. These messages rapidly shaped public attention, fostering crisis-related awareness and conservation-oriented salience. This could potentially affect electricity demand before any change in regulated tariffs occurs. Crucially, attention fluctuates on a daily basis, capturing rapid changes in exposure to information, whereas the price channel operates on a quarterly or biannual basis. This frequency differential is central to identification: estimation exploits short-term movements in attention that precede and are largely independent of effective changes in retail prices. This isolates a fast-moving behavioural response channel.

The role of price anticipation. During the crisis, the government's communication explicitly incorporated price signals, warning households about impending tariff increases and rising energy bills. Therefore, price anticipation is not a confounding factor requiring separate control, but rather a mechanism through which communication operates. The policy-relevant estimated is the total effect of government messaging on consumption adjustments, including anticipatory responses to embedded price information, precisely the outcome that crisis communication campaigns aim to achieve.

The empirical objective is therefore to estimate whether this high-frequency attention channel was operational during the crisis, and to quantify its consumption effects.

The structure of these two channels is summarised in the directed acyclic graph (DAG) presented in Figure 4. Identification is based on the backdoor criterion (Pearl et al., 2000), meaning that there is no need to include the communication measure if indeed the attention is derived from communication. There also exists a set of control elements such as temperature and holidays as they represent exogenous determinants of consumption.

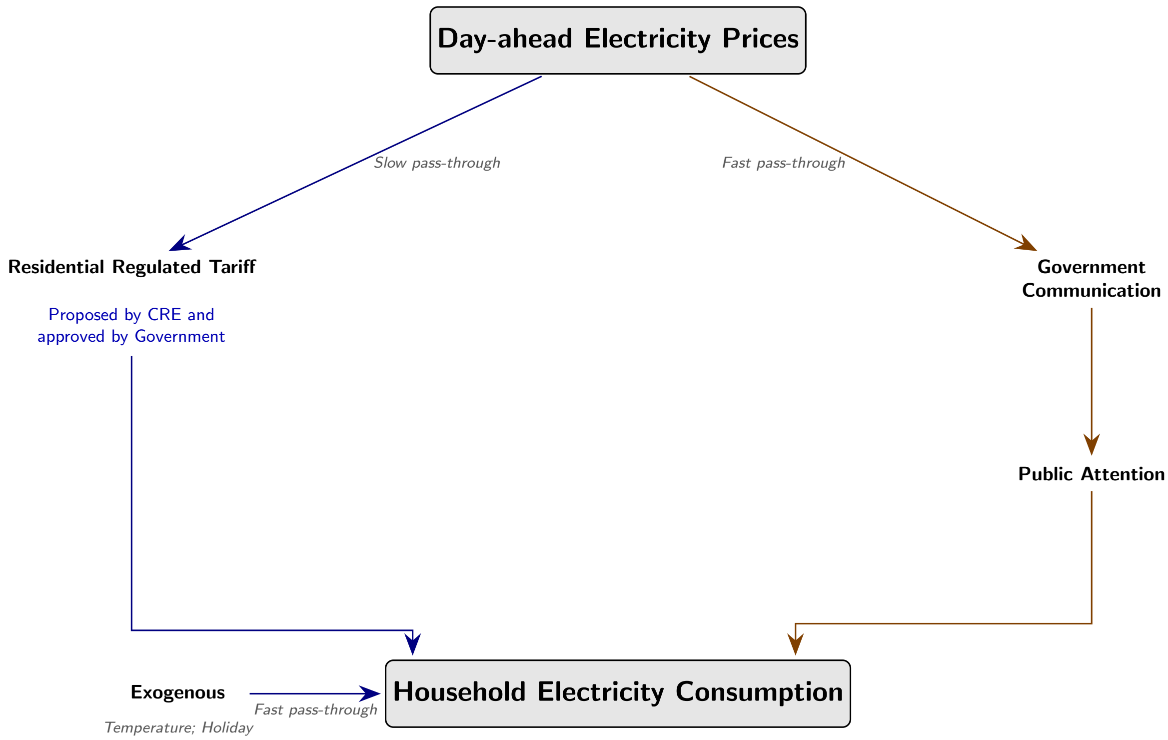


Figure 4: Directed Acyclic Graph (DAG)

5.2 Government Communication to Public Attention Identification

Before estimating the impact of attention on electricity demand, it is important to understand how government communication influences public attention. This step addresses the key challenge of identifying the effect of government communication on public awareness, as different government narratives tend to occur in close succession, making it difficult to isolate the effect of any single message type.

The empirical approach orthogonalises communication measures to recover narrative-specific variation. Let $k \in \{\text{Crisis}, \text{Conservation}, \text{Both}\}$ denote three mutually exclusive communication categories, where *Crisis* includes only statements emphasising supply risks and price volatility, *Conservation* includes only statements promoting voluntary demand reductions, and *Both* includes statements combining these themes. For each narrative k and day t, the communication variable $g_{k,t}$ is projected onto the remaining two narratives:

$$g_{k,t} = \alpha_{-k,1} g_{-k,1,t} + \alpha_{-k,2} g_{-k,2,t} + u_{k,t}, \tag{4}$$

where $-k$ indexes the other two narratives. The residual $\hat{u}_{k,t}$ represents variation in narrative k that is orthogonal to correlated messaging, isolating its unique informational content.

This orthogonalised measure is then used to predict Google Search Volume for the corresponding narrative, controlling for lagged attention and public service announcements. The procedure produces two key outputs. First, second-stage fitted values construct a narrative-induced attention measure that isolates the component of observed attention driven by government communication. Second, estimated marginal effects quantify how communication shifts attention levels.

5.3 A Restricted Error Correction Model

Based on Jamissen et al. (2024) and Loi and Loo (2016), the French residential consumption for electricity is modeled using an ARDL:

$$e_{i,t} = \alpha_i + \sum_{k=1}^{p} \gamma_{i,k} e_{i,t-k} + \sum_{j=1}^{n} \sum_{l_j=0}^{q_j} \beta_{j,i,l_j} \mathbf{x}_{j,i,t-l_j} + \epsilon_{i,t}, \tag{5}$$

where $e_{i,t}$ represents the seasonally adjusted electricity consumption for tariff profile i of the residential sector at day t, without accounting for weather variability. The model incorporates a set of explanatory variables, denoted by the vector $\mathbf{x}_{j,i,t}$. Specifically, $x_{HDD,t}$ represents heating degree days (HDD), used to quantify heating needs based on outdoor temperature.[9] Then, $x_{price,i,t}$ is the varying part of the national regulated electricity tariff, without taxes, which serves as a proxy for the consumer-paid price of electricity. Finally, $x_{m(g)_{i,t}}$ reflects a moving average in consumer attention (m) related to government incentives (g). While changes in attention are not necessarily assumed to have direct effects on electricity use, sustained or repeated exposure to these narratives may alter consumption patterns over time. Accordingly, each constructed series $m(g)_{i,t}$ is subsequently smoothed over multiple potential time horizons.

Finally, the specification incorporates a binary variable, x_{dry}, which equals one during periods of load control implemented by the grid operator. This variable captures the struc-

[9] The specification is supposed to be linear, in particular because, as emphasised by Jamissen et al. (2024), the transformation of raw temperature into HDD already embeds the nonlinearity of heating needs. The remaining relationship between HDD and energy demand is well approximated by a linear form.

tural reduction of discretionary demand when hot-water systems are remotely curtailed. In practice, x_{dry} enters both as a main effect and through interaction with the attention variables. The combined coefficients of each pair (e.g. conservation attention and its load control interaction) are then tested jointly, allowing the estimation of communication effects separately during and outside of load-control periods. This specification ensures that the interpretation of attention-induced demand adjustments reflects the institutional constraint of reduced flexibility in load control periods, while still capturing their potential amplifying effect in normal conditions.[10]

As control variables, but not interpreted in the results, the model also integrates a binary indicator for lockdowns during COVID-19, for holidays and for the subscription part of the pricing scheme, following Auray et al. (2019). The model residuals, $\epsilon_{i,t}$, are assumed to follow a centered normal distribution, $\epsilon_{i,t} \sim N(0, \sigma^2_{\epsilon_{i,t}})$. The lag parameters p and q are selected by minimising the Akaike Information Criterion (AIC).

Beyond this initial specification, the ARDL model, if cointegration exists, can be reframed as a Restricted Error Correction Model (RECM):

$$\Delta e_{i,t} = \underbrace{\phi_i \left(e_{i,t-1} - \sum_{j=1}^{n} \theta_{i,j} x_{j,i,t-1} \right)}_{\text{Correction towards long-term equilibrium}} + \underbrace{\sum_{k=1}^{p-1} \lambda_{i,k} \Delta e_{i,t-k}}_{\text{Inertia of demand}} + \underbrace{\sum_{j=1}^{n} \sum_{l_j=0}^{q_j} \delta_{j,i,l_j} \Delta x_{i,j,t-l_j}}_{\text{Short-term effects of } x_{j,i}} + \epsilon_{i,t}, \qquad (6)$$

with ϕ_i defined as the speed of adjustment towards the equilibrium relationship, represented in parentheses, and also called the long-term dynamics.[11] The third term, using first-order differences of independent variables, captures the short-term dynamics.

The choice of the ARDL/RECM framework is motivated by three considerations. First, it is well established in the energy demand literature (e.g. Jamissen et al., 2024; Loi and Loo, 2016), and offers a tractable single-equation alternative to VAR or VECM models. Second, its ECM representation allows for a natural distinction between short-run responsiveness of electricity demand (e.g. to weather shocks or communication campaigns) and long-run equilibrium relationships, which is particularly relevant when analysing household behaviour during crisis periods. Third, it is particularly flexible with respect to the order of integration: RECM estimations remain valid when regressors are a mixture of $I(0)$ and $I(1)$ series and accommodates heterogeneous lag structures across regressors, enabling the model to capture thermal inertia in heating and behavioural frictions in price or communication responses.[12]

[10] See Appendix B.3 for the exact formulation of combined effects and their standard errors.
[11] For convergence, ϕ_i must be negative, significant, and less than unity in amplitude.
[12] Cointegration is tested using Pesaran et al. (2001) bounds testing, which evaluates whether the adjust-

6 Results

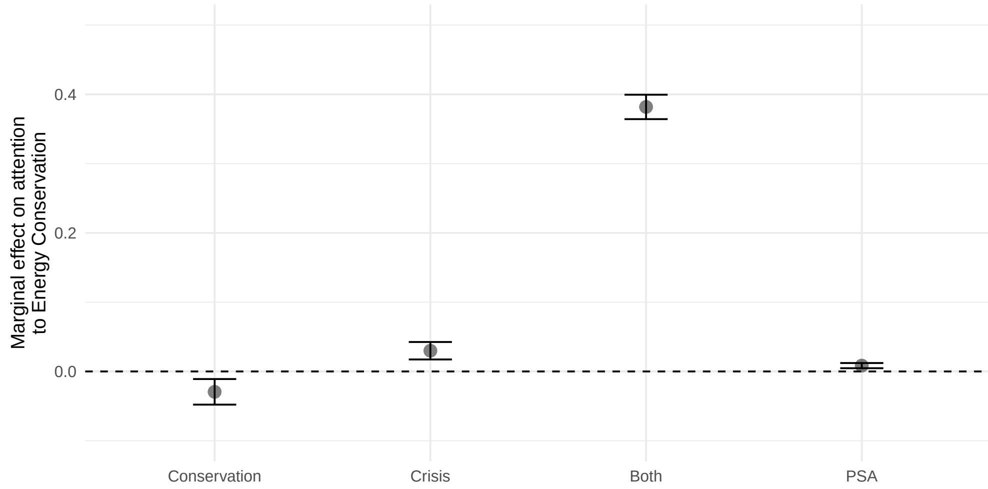
This section presents the empirical findings in two stages. First, estimated marginal effects quantify the impact of government communication on public attention to energy narratives. Secondly, the ARDL-RECM framework estimates estimated short-run and long-run responses of residential electricity consumption to key drivers. Short-run coefficients capture immediate responses to one-time shocks, while long-run coefficients represent the steady-state equilibrium relationships towards which daily consumption converges following sustained changes in prices, temperature or attention. The demand analysis distinguishes between two representative household tariff profiles: base pricing and time-of-use (ToU) pricing.

6.1 From Communication to Attention

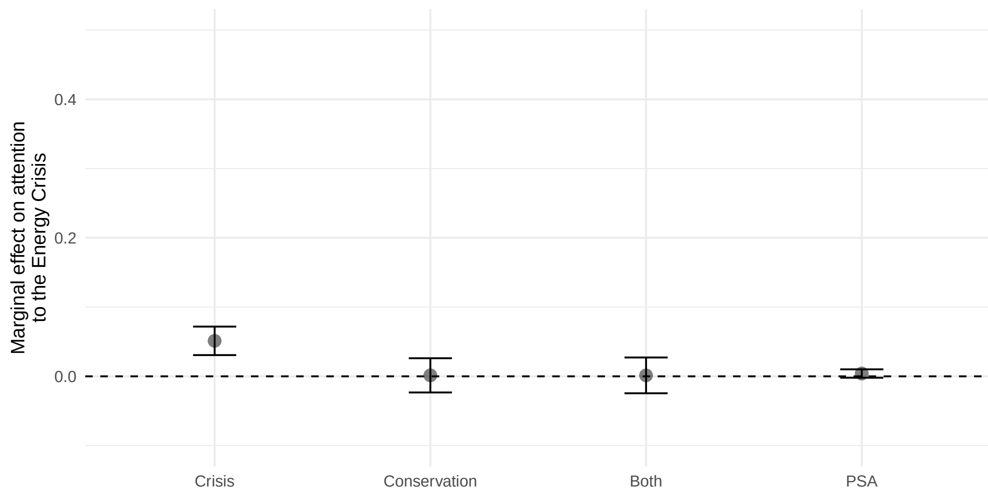
Attention to energy conservation only increases significantly (38%) when conservation and crisis messages are communicated together, combining economic concerns with actionable conservation responses. Public service announcements generate minimal responses while isolated conservation appeals marginally decrease the attention for the topic. In contrast, attention to the energy crisis is driven exclusively by very specific crisis communication. Notably, this crisis attention remained subdued during October–November 2022 despite active government messaging, probably because all of the attention is translated trhough the conservation attention that is a joint reaction to price signal and conservation incentives. The crisis attention itself only surged from 6 December 2022 onwards when communication shifted towards power failure warnings rather than price volatility. This results aligns with behavioural economics evidence suggesting that urgency and physical salience are prerequisites for effective information-based interventions (DellaVigna and Gentzkow, 2010; Blasco and Gangl, 2023; Bolderdijk et al., 2013).

The two time series of attention used for the rest of the analysis therefore capture different household concerns: attention to conservation reflects economic concerns paired with behavioural guidance, translating crisis attention into actionable responses, while attention to the crisis reflects fear of tangible supply disruption; households paid particular attention when blackouts became a real threat.

ment coefficient ϕ_i differs significantly from zero. Unit root tests confirm price and crisis attention are $I(1)$ while other series are $I(0)$ (Appendix B, Table 9), making ARDL preferable to methods requiring all $I(1)$ series (Engle and Granger, 1987; Johansen, 1995). Test statistics exceed upper critical bounds for both tariff types (Appendix B, Table 10), confirming cointegration and validating the RECM specification.



(a) Attention to Energy Conservation



(b) Attention to Energy Crisis

Figure 5: Estimated effects of Government Communication on Public Attention

6.2 Short-run dynamics

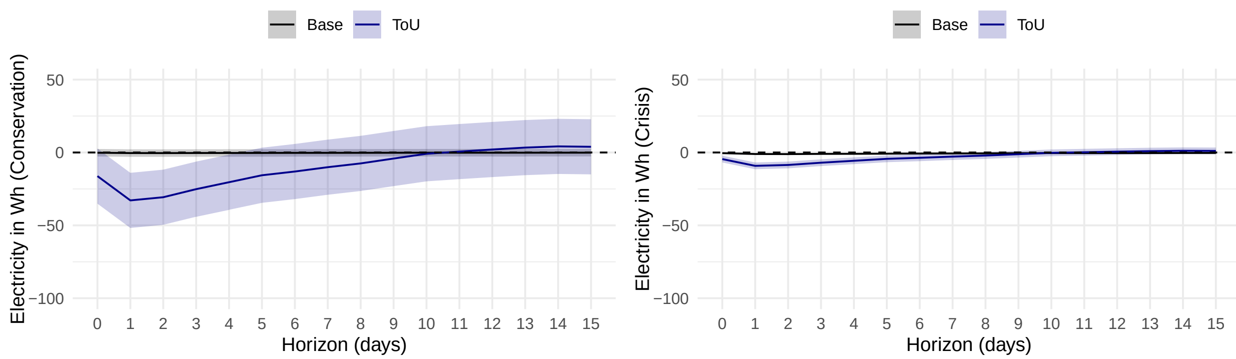
Attention Delay Throughout the estimation process, the lag structure between cumulative attention and consumption changes is optimised to identify the temporal window over which attention exposure influences behavior. For conservation-related attention, the optimal moving average window is 90 days, indicating that households adjust consumption only after sustained and repeated exposure. This gradual response is consistent with behavioural adjustment through habit formation and norm internalisation (Allcott and Rogers, 2014), whereby conservation requires persistent cognitive engagement before translating into durable practice. In stark contrast, crisis-related attention exhibits an optimal window of only 0 to 1 day, revealing that households react instantaneously once communications signal imminent power failure risks.

Consequently, the following estimates represent variations in electricity consumption arising from two distinct attention mechanisms: cumulative conservation messaging accumulated over 90 days, and immediate crisis alerts triggering same-day responses to potential blackout threats.

Attention to Conservation For ToU households, conservation attention yields an estimated elasticity of 0.0005 on the day of the shock (Panel A, Table 4), accompanied by a very wide confidence interval. This substantial uncertainty reflects potential heterogeneity in behavioural responses that cannot be fully captured by the representative household framework employed here. Households exhibit varying degrees of responsiveness to conservation messaging, likely driven by differences in environmental preferences, media attentiveness, or capacity to adjust consumption patterns. Figure 6 traces the dynamic adjustment path over the subsequent 15 days, revealing a peak reduction of approximately 30 Wh before consumption returns to equilibrium by day 4. This relatively rapid stabilisation suggests that while conservation messages initially trigger behavioural change, the effect dissipates quickly in the absence of sustained reinforcement.

Attention to Crisis Crisis attention produces a markedly different response pattern. For ToU households, the estimated elasticity is 0.0002 on the day of the shock (Panel A, Table 4), with a substantially narrower confidence interval than observed for conservation attention. This tighter precision indicates greater homogeneity in behavioural responses to crisis communications, suggesting that the threat of service interruption triggers more uniform reactions across households than voluntary conservation appeals. Figure 6 shows that crisis attention generates a maximum reduction of approximately 9 Wh, less than one-third the magnitude of the conservation effect, with equilibrium restored by day 6. The more modest response magnitude, despite higher behavioural consistency, likely reflects the fact that crisis-driven curtailment targets only discretionary loads that can be shed immediately, whereas conservation messaging may prompt broader adjustments.

Base tariff households present a distinctive pattern. While Figure 6 does not clearly illustrate their response trajectory, Table 4 reveals that these households also exhibit a statistically significant response to crisis attention. However, the point estimate approaches zero in terms of Wh consumption reduction, suggesting that base households possess less flexibility to curtail consumption in response to crisis signals.



Notes : The delay represents, on each day following the initial shock, the short-run impact in kWh/contract. Once the delay returns to equilibrium, it indicates that the shock has been fully absorbed by the system. The solid line represents the estimated coefficients and the shaded area represents the interval confidence at 95%.

Figure 6: Short-Run dynamic estimates

6.3 Long-run Equilibrium

Price Elasticities The long-run price elasticities estimated from the ARDL-RECM framework reveal important insights into household responsiveness during an unprecedented energy crisis. Base tariff households exhibit a long-run price elasticity of 0.16, while ToU households display a somewhat higher elasticity of 0.21 (Table 4). The modestly higher elasticity observed for ToU households suggests that tariff structure is a proxy for more flexible type of household. Nonetheless, both elasticities are below literature estimations.

There is a well-established consensus that energy demand is relatively price-inelastic, with a estimates below one but significant (Hanemann et al., 2013; Alberini et al., 2019; Frondel et al., 2019; Ewald et al., 2021). For residential electricity under normal market conditions, the literature reports elasticities ranging from 0.25 (Pellini, 2021) to 0.80 (Auray et al., 2019). This paper elasticities fall substantially below this range, aligning instead with emerging evidence on residential energy demand during the recent crisis, accounting for both monetary and non-monetary incentives. Studies examining natural gas consumption in Germany residential sector during the same period report similarly compressed elasticities: 0.16 (Ruhnau et al., 2023), 0.03 (Jamissen et al., 2024) and 0.07 (Behr et al., 2025) . These results are also in line with earlier works such as Reiss and White (2008) for California and Ito et al. (2018) for Japan. These estimates indicate that not accounting for crisis-linked non-monetary motivations in estimation strategies may lead to an overestimation of the price elasticity of energy by falsely attributing non-price-driven savings to price-induced savings.

Attention Elasticities For conservation-related attention, base tariff households exhibit an elasticity of 0.0005, while ToU households show a substantially larger elasticity of 0.0050, though neither estimate reaches statistical significance in the baseline specification (Panel B, Table 4). Crisis-related attention generates more robust and uniform responses: both base and ToU households display a statistically significant elasticity of 0.0016, indicating that blackout risk messaging produces consistent long-run demand adjustments regardless of tariff structure.

Translating these elasticities into tangible daily consumption impacts provides practical interpretation. A one-unit increase in crisis attention yields a long-run reduction of 46.9 Wh (-0.3%) per day for ToU households and 22.1 Wh (-0.5%) per day for base households. Conservation attention generates larger but more uncertain effects: 168.3 Wh (-1.2%) per day for ToU households and only 8.3 Wh (-0.2%) per day for base households, with the base estimate remaining statistically insignificant due to potential lack of flexibility.

Table 4: Short and Long-Run Estimates - RECM

	Short-Run				Long-Run			
	Base		ToU		Base		ToU	
Variable	Elast.	Wh/day	Elast.	Wh/day	Elast.	Wh/day	Elast.	Wh/day
Panel A. Price and Temperature								
Intercept		91.0^{***}		$1,371^{***}$		$4,459^{***}$		$14,236^{***}$
		(28.8)		(206)		(939)		(1,610)
Price	0.0461^{***}	-3.1^{***}	0.1804^{***}	-28.0^{***}	0.1624^{***}	-10.9^{***}	0.2112^{***}	-32.8^{***}
	(0.0133)	(0.9)	(0.0500)	(7.8)	(0.0459)	(3.1)	(0.0347)	(5.4)
Temperature	0.0149^{***}	28.1^{***}	0.0750^{***}	315.5^{***}	0.0042	8.0	0.2353^{***}	989.6^{***}
	(0.0007)	(1.2)	(0.0021)	(8.1)	(0.0323)	(60.9)	(0.0166)	(68.7)
Panel B. Attention Effects								
Conservation	0.0000	-1.4	0.0010	-63.7	0.0023	-67.3	0.0102	-661.0
	(0.0006)	(18.1)	(0.0020)	(131.7)	(0.0305)	(888.1)	(0.0211)	(1,368)
Crisis	0.0000	-0.8	0.0003	-16.4	0.0015	-38.9	0.0030	-170.3
	(0.0001)	(3.0)	(0.0004)	(21.6)	(0.0056)	(144.8)	(0.0041)	(224.0)
Panel C. Load Control Effects								
Load control	0.0000	-0.8	0.0010^{***}	-82.6^{***}	0.0010	-37.8	0.0102^{***}	-857.8^{***}
	(0.0001)	(4.0)	(0.0004)	(30.0)	(0.0052)	(196.7)	(0.0036)	(299.3)
Conservation	0.0000	1.2	0.0007	47.4	0.0018	59.0	0.0068	492.7
	(0.0006)	(18.4)	(0.0019)	(134.1)	(0.0279)	(904.4)	(0.0192)	(1,393)
Crisis	0.0000	0.3	0.0002	11.9	0.0006	16.9	0.0020	123.4
	(0.0001)	(3.0)	(0.0004)	(21.7)	(0.0052)	(145.2)	(0.0037)	(224.4)
Panel D. Combined Effects								
Conservation	0.0000	-0.2	0.0005	-16.2	0.0005	-8.3	0.0049	-168.3
	(0.0001)	(1.6)	(0.0003)	(11.5)	(0.0050)	(76.7)	(0.0035)	(118.8)
Crisis	0.0000	-0.5^{**}	0.0002^{**}	-4.5^{***}	0.0016	-22.1^{**}	0.0016^{**}	-46.9^{***}
	(0.0000)	(0.2)	(0.0001)	(1.4)	(0.0010)	(10.8)	(0.0008)	(15.4)
R^2	0.992		0.992		0.992		0.986	
Observations	1,839		1,839		1,839		1,839	

Notes : Elasticities represent the ratio of proportional changes in quantity demanded to proportional changes in independent variables; values below 1 in absolute value indicate inelastic demand. Estimates are expressed in Wh/day/contract. Data span July 1, 2019 to June 30, 2024. *** p<0.01, ** p<0.05, * p<0.1.

Electricity Decrease Decomposition To synthesise the long-run estimates and quantify their practical significance, Figure 7 presents a decomposition of the observed demand reduction for aggregation at the national level. Overall, electricity consumption in the residential section declined by 13 TWh between the pre-crisis baseline period (2019-21) and the crisis period (2022-23), falling from 149 TWh to 136 TWh.

The econometric estimates allow us to decompose this reduction into its constituent drivers. Price increases account for the largest share, contributing 6.9 TWh (54% of the total decrease). This reflects the effect of regulated tariff adjustments that raised residential electricity costs to unprecedented levels, even considering the tariff shield. Temperature effects explain 2.2 TWh (17%), attributable to milder winter conditions during 2022-23 that reduced heating-related electricity demand. Load control measures implemented by the public distribution network operator contributed to a similar 2.4 TWh (18%). Beyond these physical and price-driven mechanisms, government communications generated measurable demand reductions through behavioural channels. Conservation-related messaging, appeals to voluntary energy savings framed around environmental responsibility and collective effort, induced a reduction of 1.1 TWh (8%), while crisis-related communications warning of blackout risks contributed 0.4 TWh (3%). Collectively, attention-driven behavioural adjustments account for approximately 11% of the total decrease, a non-negligible share achieved through information provision alone. However, it does only represent a decrease in national consumption of 1% between 2019-2021 and 2022-2023.

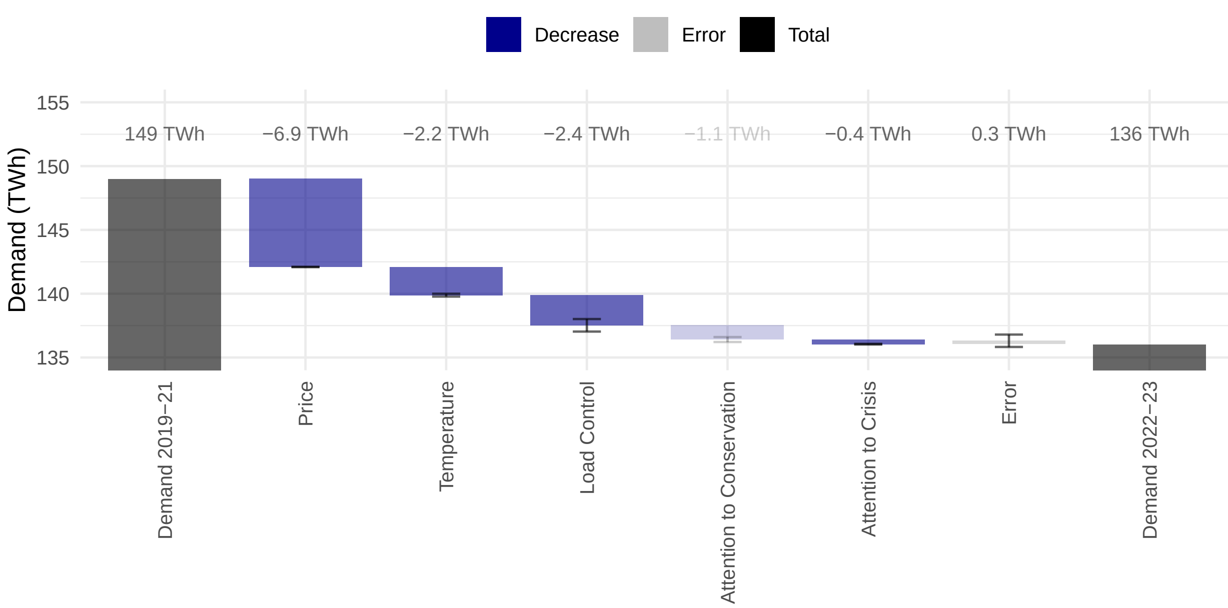


Figure 7: Electricity Decrease Decomposition

6.4 Robustness

Censored Sample It could be argued that these estimates are affected by the load control measure, which automatically reduce electricity demand for ToU type of household. Since this system mechanically reduce consumption during midday period, households have limited flexibility to voluntarily reduce usage in response to attention. In order to isolate true behavioural flexibility among households, the model was re-estimated, excluding consumption during load control hours (11am – 2pm). Details are provided in Appendix C.3.

Table 5 presents the estimates for the censored sample excluding load control hours. The most substantial change occurs in the conservation attention elasticity, which becomes strongly statistically significant, with a decrease of 255 Wh/day (−2.25%). This shift suggests that the baseline specification masked a robust conservation response by averaging together hours when households could voluntarily adjust consumption and hours when load control mechanically constrained their average behavior. ToU households thus possess genuine and substantial flexibility to respond to conservation messaging when not technically constrained by grid-operator interventions.

Table 5: Censored Sample - RECM

	Short-Run				Long-Run			
	Base		ToU		Base		ToU	
Variable	Elast.	Wh/day	Elast.	Wh/day	Elast.	Wh/day	Elast.	Wh/day
Panel A. Price and Temperature								
Intercept		55.0***		1,098***		3,177***		11,294***
		(18.2)		(160)		(719)		(1,208)
Price	0.0440***	−2.2***	0.1793***	−21.4***	0.1640***	−8.1***	0.2436***	−29.1***
	(0.0122)	(0.6)	(0.0500)	(6.0)	(0.0435)	(2.1)	(0.0277)	(3.3)
Temperature	0.0153***	21.0***	0.0800***	258.9***	0.0012	−1.6	0.2523***	816.3***
	(0.0006)	(0.8)	(0.0020)	(6.3)	(0.0375)	(50.9)	(0.0162)	(52.4)
Panel B. Attention Effects								
Conservation	0.0000	−0.2	0.0005***	−24.8***	0.0005	−11.6	0.0051***	−255.5***
	(0.0003)	(0.8)	(0.0001)	(7.3)	(0.0020)	(48.2)	(0.0014)	(71.4)
Crisis	0.0000**	−0.3**	0.0001***	−3.4***	0.0010**	−19.0**	0.0008***	−34.7***
	(0.0000)	(0.1)	(0.0000)	(1.1)	(0.0005)	(8.9)	(0.0003)	(11.6)
R^2	0.994		0.994		0.986		0.986	
Observations	1,839		1,839		1,839		1,839	

Notes : Elasticities represent the ratio of proportional changes in quantity demanded to proportional changes in independent variables; values below 1 in absolute value indicate inelastic demand. Estimates are expressed in Wh/day/contract. Data span July 1, 2019 to June 30, 2024. *** p<0.01, ** p<0.05, * p<0.1.

Heterogeneity by Contracted Power To explore variation in attention responsiveness, Table 6, disaggregates long-run ToU effects by contracted power capacity, a technical parameter determining maximum instantaneous load and serving as a proxy for dwelling size, heating intensity, and adjustment flexibility. Descriptive evidence reported in Appendix C.4 Table 16 highlights strong structural differences across contract levels. Higher contracted power is closely associated with larger dwellings and more energy-intensive housing types: for instance, 12 kVA contracts are exclusively found in detached houses and predominantly in dwellings above 80 m^2, while 6 kVA contracts are concentrated in apartments and smaller surfaces. A key distinction also comes from higher contracted power level household being more likely to have at least two children under 11 years old. These differences translate into substantial variation in baseline consumption, ranging from 2.3 MWh per year for 6 kVA base households to 10.3 MWh for 12 kVA ToU households. Beyond total consumption, contracted power also captures heterogeneity in end-use composition. Lower-capacity households allocate a larger share of electricity to non-heating uses such as hot water and appliances, while higher-capacity households devote a dominant share to electric heating, which accounts for more than 60% of consumption in the 12 kVA ToU group. This distinction is central for understanding flexibility: heating loads are both the main driver of peak demand and the primary margin along which consumption can be reduced or shifted.

Price and crisis elasticities remain stable across contract sizes, indicating that responsiveness to economic incentives does not scale with dwelling characteristics. Conservation attention, however, exhibits pronounced heterogeneity inversely related to contracted power. The 6 kVA segment displays the largest elasticity, corresponding to 151 Wh daily reductions (-1.25%), while higher-capacity households show insignificant elasticities despite larger absolute saving potential, 12 kVA households could achieve up to 278 Wh reductions (-1.33%). These results reflect crowding effects: high-capacity households already face binding technical constraints because of the load measure, that reduce their demand by 1 440 Wh (-6.9%), and potential thermal comfort trade-offs (e.g. young children), leaving limited margins for voluntary behavioral adjustments despite larger absolute consumption levels.

Table 6: Contracted Power - RECM

	ToU - 6 kVA		ToU - 9 kVA		ToU - 12 kVA	
Variable	Elast.	Wh/day	Elast.	Wh/day	Elast.	Wh/day
Panel A. Price and Temperature						
Intercept		$12,001^{***}$		$14,771^{***}$		$20,828^{***}$
		(1,350)		(2,121)		(1,885)
Price	0.1869^{***}	-18.7^{***}	0.1883^{***}	-32.3^{***}	0.1898^{***}	-45.2^{***}
	(0.0294)	(2.9)	(0.0359)	(6.1)	(0.0407)	(9.7)
Temperature	0.2307^{***}	623.9^{***}	0.2466^{***}	$1,143.4^{***}$	0.2236^{***}	$1,441.5^{***}$
	(0.0384)	(38.4)	(0.0785)	(78.5)	(0.1206)	(120.6)
Panel B. Attention Effects						
Conservation	0.0194	−811.5	0.0071	−512.2	0.0089	−889.0
	(0.0767)	(766.5)	(0.1561)	(1,560.9)	(0.2386)	(2,386.0)
Crisis	0.0018	−67.8	0.0034	−218.3	0.0036	−319.8
	(0.0013)	(125.6)	(0.0026)	(255.7)	(0.0039)	(390.1)
Panel C. Load Control Effects						
Load control	0.0093^{***}	-506.5^{***}	0.0100^{***}	-931.7^{***}	0.0111^{***}	$-1,440.9^{***}$
	(0.0031)	(167.8)	(0.0037)	(341.7)	(0.0052)	(521.6)
Conservation	0.0142	660.5	0.0045	361.7	0.0055	611.3
	(0.0781)	(780.7)	(0.1590)	(1,589.6)	(0.2428)	(2,428.2)
Crisis	0.0010	40.2	0.0024	163.8	0.0026	245.8
	(0.0013)	(125.9)	(0.0026)	(256.3)	(0.0039)	(390.9)
Panel D. Combined Effects						
Conservation	0.0069^{**}	-151.0^{**}	0.0040	−150.6	0.0053	−277.6
	(0.0067)	(67.1)	(0.0136)	(135.7)	(0.0208)	(208.2)
Crisis	0.0014^{***}	-27.6^{***}	0.0017^{***}	-54.5^{***}	0.0016^{***}	-74.1^{***}
	(0.0009)	(8.6)	(0.0018)	(17.6)	(0.0027)	(26.8)
R^2	0.983		0.985		0.987	
Observations	1,839		1,839		1,839	

Notes : Elasticities represent the ratio of proportional changes in quantity demanded to proportional changes in independent variables; values below 1 in absolute value indicate inelastic demand. Estimates are expressed in Wh/day/contract. Data span July 1, 2019 to June 30, 2024. *** p<0.01, ** p<0.05, * p<0.1.

7 Conclusion

This paper examined whether government communication could meaningfully influence household electricity demand during an energy crisis. This question arose from France's experience in winter 2022-23, when reduced nuclear availability and soaring wholesale prices prompted one of Europe's largest conservation campaigns. Over several months, the French authorities issued official statements urging citizens to reduce their consumption. Initially, they appealed to environmental responsibility while passing-though price signal; later, they warned of potential blackouts. Whether these appeals resulted in a measurable reduction in demand remained an open empirical question. In order to answer this question, the study traced the full causal chain from government messaging to household attention, and from attention to electricity use. Three main findings emerged. First, communication effectiveness fundamentally depends on framing. Conservation messaging only gained traction when it was embedded within broader narratives emphasising the economic consequences. Crisis communication about concrete warnings of imminent blackouts in early December 2022 drew immediate public engagement.

Secondly, attention only translates into a demand response when households possess actionable flexibility. While crisis-related attention produced relatively uniform but modest reductions across households, conservation-related attention exhibited substantial variation depending on the level of flexibility available to households. Among base tariff households, overall conservation responsiveness remains limited, linked to the initial consumption level of these type of contracts. Among time-of-use tariff households, smaller contracts demonstrate the strongest and most precisely estimated conservation response. In contrast, larger contracts exhibit weaker response because of a crowding-out effect: because of the load control measure, these households were unable to further reduce their electricity demand significantly during the targeted periods.

Finally, prices and temperatures quantitatively dominate demand responses. While communication contributed measurably during moments of acute crisis, its impact was modest compared to the ongoing influence of economic incentives and climatic conditions.

These findings are subject to several limitations. Firstly, this analysis quantifies the demand response attributable to government communication, but does not evaluate its cost-effectiveness in comparison to alternative policy instruments. Future research could compare the effectiveness of communication-based policies with that of subsidy-based policies. Secondly, the aggregate tariff-level analysis cannot capture differences in household income. Using micro-level data, future research could further decompose the effects of attention by socioeconomic status and assess whether information-based demand management exacerbates or mitigates energy poverty during crisis periods.

Stepping back, words can indeed save watts, but this paper shows that information alone is not a sufficient lever for demand-side adjustment. While government communication can trigger measurable reductions in electricity use, its effectiveness is fundamentally dependent on the availability of underlying flexibility. Households differ in both their attention to public signals and their capacity to act upon them. Consequently, identical messages generate different responses, reflecting disparities in equipment, constraints, and baseline consumption levels. These findings suggest that communication policies are most effective when combined with structural measures that increase households' ability to adjust their demand. In the absence of such measures, even the most salient messages will have limited aggregate impact and uneven distributional effects.

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A Classification Methodology for Government Communication

This appendix details the methodology used to classify government communication regarding energy conservation incentives. The classification follows four main steps: (A) textual preprocessing, (B) semi-supervised labeling, (C) training of a machine learning model using XGBoost, and (D) evaluation of the model's performance.

Step A — Textual Preprocessing

The initial corpus consists of 12,184 political communication extracted from the governmental platform Vie Publique. To prepare the corpus for classification:

- Texts were processed using `SpaCy` (French model), with removal of special characters, stopwords, conversion to lowercase, and lemmatization.
- Texts were transformed into numerical feature vectors using a `CountVectorizer`, followed by a `TfidfTransformer` to enhance informative words.

Step B — Semi-Supervised Labeling and Dataset Construction

communication were labeled semi-supervised as follows:

- **Label 1**: explicit mentions of *sobriété énergétique* or *crise énergétique*.
- **Label 0**: communication clearly focused on unrelated topics (e.g., *jeux olympiques*, *élections*, *violence*, *vaccination*).
- **Label -1**: communication for which no label could be automatically assigned.

Table 7: Distribution of Labeled Data into Training and Testing Sets

Label	Train Set	Test Set
1 (Relevant)	111	51
0 (Non-relevant)	1914	837
-1 (Unlabeled)	9267	-

Given the small size of the initially labeled dataset, a semi-supervised learning strategy was employed to expand the training data iteratively. The general principle is illustrated in Figure 8. Semi-supervised learning was performed by iteratively adding the 10 most confidently predicted examples from the unlabeled set and retraining until no confident examples remained.

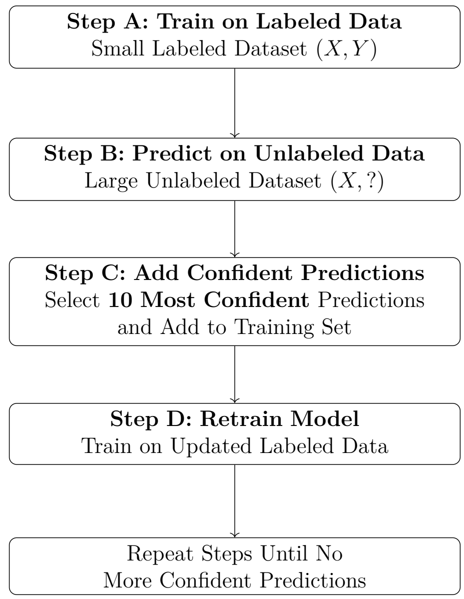


Figure 8: How is a semi-supervised algorithm trained ?

Step C — Training an XGBoost Classifier

A gradient boosting model (XGBoost) was trained, defined by:

$$P(\hat{y}_i = 1|X) = \sigma\left(\sum_{t=1}^{T} f_t(x_i)\right), \quad f_t \in \mathcal{F} \tag{7}$$

where $\sigma(z) = 1/(1+e^{-z})$ is the sigmoid activation function, f_t represents each sequentially learned decision tree, and T is the total number of boosting rounds.

The loss minimised at each iteration is:

$$\mathcal{L}^{(t)} = \sum_i L(y_i, \hat{y}_i^{(t-1)}) + \sum_t \Omega(f_t) \tag{8}$$

with:

- Cross-entropy loss $L(y_i, \hat{y}_i^{(t-1)})$ adjusted for class imbalance via ω_i:

$$L(y_i, \hat{y}_i^{(t-1)}) = -\omega_i \left[y_i \log(\hat{y}_i^{(t-1)}) + (1 - y_i) \log(1 - \hat{y}_i^{(t-1)})\right]$$

- Regularization term $\Omega(f_t)$ penalizing tree complexity:

$$\Omega(f_t) = \gamma K + \frac{1}{2}\lambda \sum_{j=1}^{K} w_j^2$$

Step D — Evaluation of Classification Performance

Model evaluation was conducted on the test set, showing high predictive performance for both topics. For the detection of *sobriété énergétique* communication, the model achieved an accuracy of 99.5%, a precision of 98.0%, a recall of 94.1%, and an F1 score of 96.0%. For the detection of *crise énergétique* communication, the model achieved an accuracy of 99.6%, a precision of 93.4%, a perfect recall of 100%, and an F1 score of 96.6%. These results indicate that the classification process reliably identifies relevant government communication while minimizing false positives and false negatives.

Table 8: Classification Performance Summary

	Sobriété énergétique		**Crise énergétique**	
	True 0	True 1	True 0	True 1
Predicted 0	836	3	859	0.0
Predicted 1	1	48	4	57
Accuracy	0.995		0.996	
Precision	0.980		0.934	
Recall	0.941		1.000	
F1 Score	0.960		0.966	

Notes: Confusion matrices report counts of correctly and incorrectly classified communication. Metrics are computed on the fully labeled test set.

Discussion

While the classification procedure achieves strong predictive performance, with high accuracy and recall, some limitations must be acknowledged. The initial labeling process relied on specific keywords, potentially reducing the model's ability to capture more implicit or nuanced references to energy conservation. Although the semi-supervised expansion helps mitigate this constraint by enlarging the training set, the final classification remains partly conditioned by the original labeling choices. These limitations are unlikely to significantly affect the detection of explicit government calls for conservation, which are the primary interest of this study. Future improvements could involve replacing the TF-IDF vectorization with word embeddings from pre-trained language models such as CamemBERT. Embedding-based approaches would better capture semantic nuances and context, enhancing the model's ability to classify more subtle or indirect references to energy conservation.

Moving Average Measures

Relying solely on the raw daily count of classified communication would probably be insufficient to capture the real behavioural stimulus, as it is unlikely that a single isolated statement triggers an immediate and significant response in residential attention to the narratives (Allcott and Rogers, 2014). Thus, an indicator is constructed to proxy for the *buzz* given to energy savings relative to the total political communication. Similar approaches are found in the monetary policy literature, where the effectiveness of ECB communication has been shown to depend not just on the number of announcements, but also on their intensity, prominence, and cumulative visibility. For example, Istrefi et al. (2025) documents that inter-meeting speeches and interviews by ECB Governing Council members move markets nearly as much as formal policy announcements, underlining the importance of communication volume and timing. Likewise, Jarociński and Karadi (2020) shows that the market impact of ECB communication depends on the frequency and clustering of messages, while Berger et al. (2006) develops a systematic index of communication tone to capture how repeated and salient statements shape expectations. These studies motivate the use of "buzz" metrics to capture the cumulative and amplifying effects of political communication on public attention.

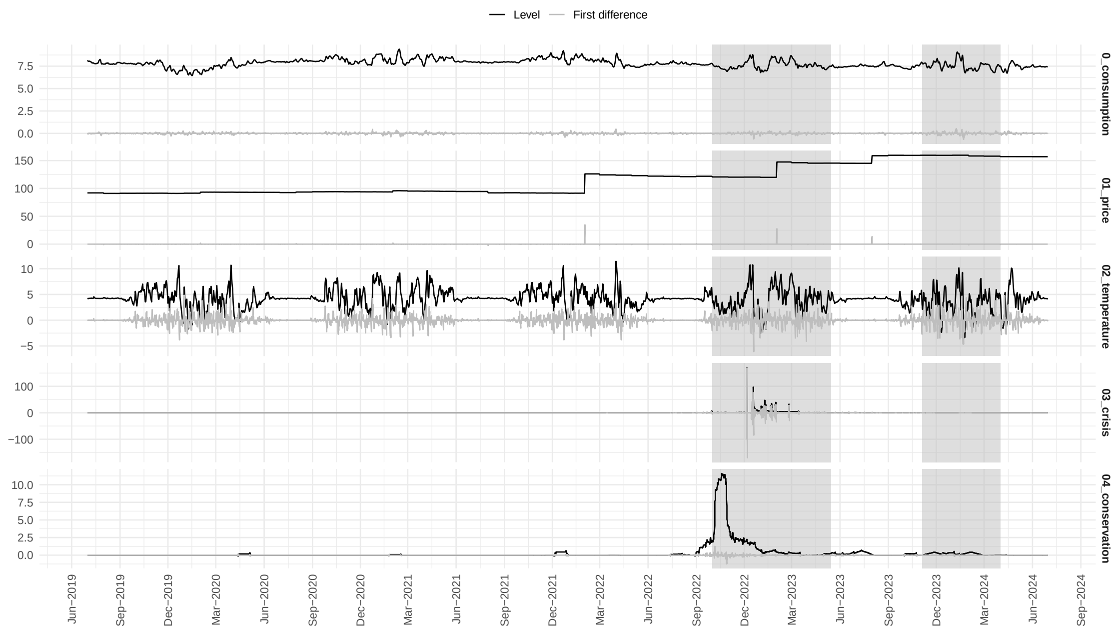
This paper "buzz-weighted" metric (g_{kt}) is defined in equation 9.

$$(g_{k,t,n}) = \underbrace{\frac{1}{n}\sum_{0}^{n-1}}_{\text{Cumulative}} \underbrace{\left(\frac{1}{N_{t-j}}\sum_{i=1}^{N_{t-j}} \omega_i g_{kti}\right)}_{\text{Buzz}} \tag{9}$$

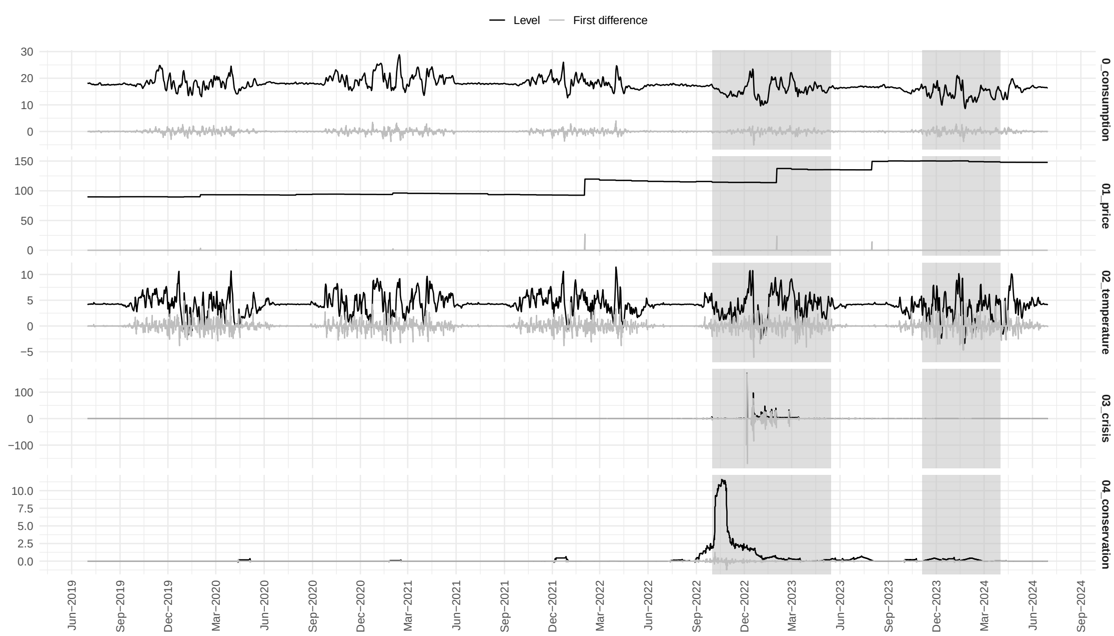
With $g_{k,t,n}$ representing the presence of topic k in the i-th statement on day t, N_t is the total number of communications on day t, and ω_i is a weight reflecting the prominence of the statement. To construct a robust index of government communication, alternative specifications are generated by combining moving-average windows with different weighting schemes. Eleven windows, ranging from 1 to 90 days, are applied to three metrics: the raw count of messages, a buzz index with uniform weights, and a buzz index with weights based on whether the term appears in the title or the type of communication (e.g., interview). This yields 33 candidate measures, from which the optimal specification is selected using the Akaike Information Criterion (AIC), balancing fit and parsimony.

B ARDL-RECM

B.1 Variables



(a) Base



(b) ToU

Figure 9: Level and First Difference Main Regressors

B.2 Co-Integration

Table 9: Unit-Roots Tests

	Level - $I(0)$				First Difference -$I(1)$			
		ADF	PP	KPSS		ADF	PP	KPSS
	Lags	*Statistics*	*P-value*	*P-value*	*Lags*	*Statistics*	*P-value*	*P-value*
e_t	9	-7.97***	0.01	0.01	8	-16.69***	0.01	0.1
x_{HDD}	4	-12.22***	0.01	0.10	5	-22.94***	0.01	0.1
x_{price}	1	-0.44	0.49	0.01	1	-30.33***	0.01	0.1
$x_{m(g)_{con}}$	8	-3.51	0.48	0.02	10	-10.50***	0.01	0.1
$x_{m(g)_{crisis}}$	10	-5.34***	0.01	0.01	10	-26.09***	0.01	0.1
x_{dry}	1	-2.45	0.24	0.01	1	-30.27***	0.01	0.1

Notes : The lag column represents the number of lags included in the ADF regression, guided by the Akaike Information Criteria.

Table 10: Co-integration bounds tests

	F-statistics	Statistic	Lower-bound $I(0)$	Upper-bound $I(1)$
Base model	6.02	-5.43***	-3.43	-4.98
ToU model	22.73	-12.29***	-3.43	-4.98

Notes : Critical bounds are provided at 1% significance level

B.3 Computation of Combined Effects During Load-Control Periods

When an attention variable x_m enters both directly and through interaction with the load-control dummy x_{dry}, the specification reads as in Eq 10.

$$e_t = \ldots + \beta_m x_{m,t} + \beta_{m\times dry}\left(x_{m,t} \times x_{dry,t}\right) + \epsilon_t. \tag{10}$$

The short-run effect of x_m when $x_{dry} = 0$ is simply:

$$\hat{x}_m = \beta_m, \tag{11}$$

while during load-control periods ($x_{dry} = 1$) the effect becomes:

$$\hat{x}_{m+dry} = \beta_m + \beta_{m\times dry}. \tag{12}$$

In the long-run representation implied by the RECM, the multipliers are given by Eq 13 where ϕ is the adjustment coefficient from the error-correction term.

$$\hat{x}_m^{LR} = -\frac{\beta_m}{\phi}, \qquad \hat{x}_{m+dry}^{LR} = -\frac{\beta_m + \beta_{m\times dry}}{\phi}, \tag{13}$$

Standard errors are then obtained using the delta method. For the short-run combined effect, the variance as in Eq 14.

$$\mathrm{Var}(\beta_m + \beta_{m\times dry}) = \mathrm{Var}(\beta_m) + \mathrm{Var}(\beta_{m\times dry}) + 2\,\mathrm{Cov}(\beta_m, \beta_{m\times dry}). \tag{14}$$

For the long-run effect, the gradient vector of Eq 15 with respect to $(\beta_m, \beta_{m\times dry}, \phi)$ is applied to the variance–covariance matrix of the estimated parameters gives the variance as presented in Eq 16 where ∇g denotes the gradient of $g(\cdot)$

$$g(\beta_m, \beta_{m\times dry}, \phi) = -\frac{\beta_m + \beta_{m\times dry}}{\phi} \tag{15}$$

$$\mathrm{Var}\Big(g(\hat{\beta}_m, \hat{\beta}_{m\times dry}, \hat{\phi})\Big) = \nabla g^{\top}\,\widehat{\mathrm{Var}}(\hat{\beta}_m, \hat{\beta}_{m\times dry}, \hat{\phi})\,\nabla g, \tag{16}$$

C Complementary Empirical Strategy

C.1 Data Sources

Table 11: Summary of Data Sources and Coverage

Source	**Frequency**	**Coverage**	**Notes**
Enedis Open Data	Daily	01/2019-04/2024	95% of mainland households
Elecdom	Daily	07/2022-07/2023	100 households
ECMWF	Daily	01/2019-04/2024	Population-weighted averages
CRE	Bi-annual	01/2019-04/2024	Weighted by contract distribution
ENTSO-E	Daily	01/2019-04/2024	European Transparency Platform
Vie Publique	Daily	01/2019-04/2024	Classified into crisis and conservation narratives
Google Trends	Daily	01/2019-04/2024	Scaled 0–100; proxy for public attention

Notes : All variables are aggregated to daily frequency for consistency. Weather indicators are weighted by population across regions. Retail tariffs correspond to regulated price schedules, while wholesale prices represent day-ahead market quotations. Public communication data are manually classified by narrative using a semi-supervised algorithm.

Table 12: Yearly rate change in the number of contracts by profile (%)

periode	Base	ToU	Tempo	Total
2019-01-01				
2020-01-01	+1.65 %	+0.96 %	-5.17 %	+1.28%
2021-01-01	+2.03 %	-0.56 %	-6.53 %	+0.80%
2022-01-01	+2.28 %	-0.40 %	+0.66 %	+1.07%
2023-01-01	-0.60 %	+1.43 %	+92.05 %	+0.89%
2024-01-01	-0.63 %	+0.23 %	+100.91 %	+0.98%

Notes : Between 2020-01-01 and 2021-01-01, the basecontracts increased by 2.0%, ToU decreased by 0.6 % and Tempo decreased by 6.5%. Overall, Enedis acquired around 1% new contracts every year between 2019 and 2024.

C.2 Attention Formation

Constructing consistent daily GSV series presents technical challenges due to Google Trends' 90-day limit for daily-frequency queries and internal rescaling across time windows. A multi-step reconstruction algorithm is applied to overcome these limitations. Overlapping 90-day blocks are retrieved and stitched together by rescaling values in overlapping periods, yielding a continuous daily index. This daily series is then aligned with official weekly aggregates by computing period-specific scaling factors, ensuring consistency across temporal frequencies while preserving within-week variation.The reconstruction procedure follows Eichenauer et al. (2022). An important caveat is that GSV captures search behavior only among internet users. In the French context, this likely under-represents older households and rural populations with lower internet access rates, while over-representing younger, urban, and digitally connected demographics. The measure should therefore be interpreted as reflecting digitally expressed attention rather than universal public awareness or comprehension of energy issues.

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Table 13: Top Search Queries related to Energy Conservation and Energy Crisis

Panel A. Energy Conservation	
French	English
sobriété énergétique	energy conservation
plan de sobriété	conservation plan
la sobriété énergétique	energy conservation
la sobriété	conservation
définition sobriété	definition of conservation
sobriété énergétique gouvernement	government energy conservation
sobriété énergétique définition	definition of energy conservation
plan sobriété énergétique	energy conservation plan
plan de sobriété énergétique	energy conservation plan
Panel B. Energy Crisis	
French	English
crise énergétique	energy crisis
crise énergétique européenne france	european and french energy crisis
crise énergétique européenne	european energy crisis
crise énergétique europe	energy crisis in Europe
crise énergétique 2022	2022 energy crisis
la crise énergétique en france	energy crisis in France
la crise énergétique	the energy crisis
crise énergétique pourquoi	why the energy crisis
crise énergétique france	france energy crisis

Table 14: Marginal Attention to Communication

	Dependent variable: attention	
	(Conservation)	(Crisis)
Panel A. Main Effects		
lag(attention)	0.587***	0.704***
	(0.012)	(0.012)
u_{con}	−0.029***	
	(0.009)	
u_{crisis}		0.051***
		(0.010)
u_{joint}	0.382***	0.001
	(0.009)	(0.013)
Panel B. Additional Controls		
crisis	0.030***	
	(0.006)	
conservation		0.001
		(0.013)
tv	0.008***	0.004
	(0.002)	(0.003)
Observations	3,653	
R^2	0.714	0.513
Adjusted R^2	0.714	0.513
Residual Std. Error	1.605	2.632
F Statistic	1,821.000***	769.300***

Note: *p<0.1; **p<0.05; ***p<0.01

C.3 Robustness Without Load Control Hours

The daily consumption between 11:00 a.m. and 4:00 p.m. was set to zero to remove the direct influence of the load-control intervention, during which the grid operator remotely curtailed hot-water heating. This adjustment isolates voluntary behavioural responses by ensuring that estimated effects are not mechanically driven by enforced curtailment. A limitation is that any behavioural adaptation occurring precisely during these hours will no longer be captured. Table 15 and Figure 10 provide a descriptive comparison of electricity demand with and without these hours. Total residential consumption during the 2022-23 winter remains almost identical to the baseline, with an overall reduction of about 8% relative to pre-crisis years, smaller than the 9.3% observed when midday hours are included. The difference is concentrated among ToU households, whose consumption falls by 12.5% once load-control periods are removed, compared with 13% in the full sample. Base-tariff households show a modest decline of 3% compared to a decrease of 4.7% in the full sample.

Table 15: Average Annual Electricity Consumption - Without Load Control Hours

Panel A. Total electricity consumption (TWh)				
	2019–2021	2022–2023	Difference	%
Base	37.2	36.5	-0.7	-1.8
ToU	75.0	65.6	-9.3	-12.5
Total	112.3	102.3	-10.0	-8.9
Panel B. Average consumption per contract (MWh/contract)				
	2019–2021	2022–2023	Difference	%
Base	2.1	2.0	-0.1	-4.3
ToU	5.2	4.5	-0.7	-13.2
Total	7.3	6.5	-0.8	-10.5

Notes : Years are defined from July to July in order to encompass one winter period per year of study. Thus, the year 2022 goes from July 2022 to June 2023.

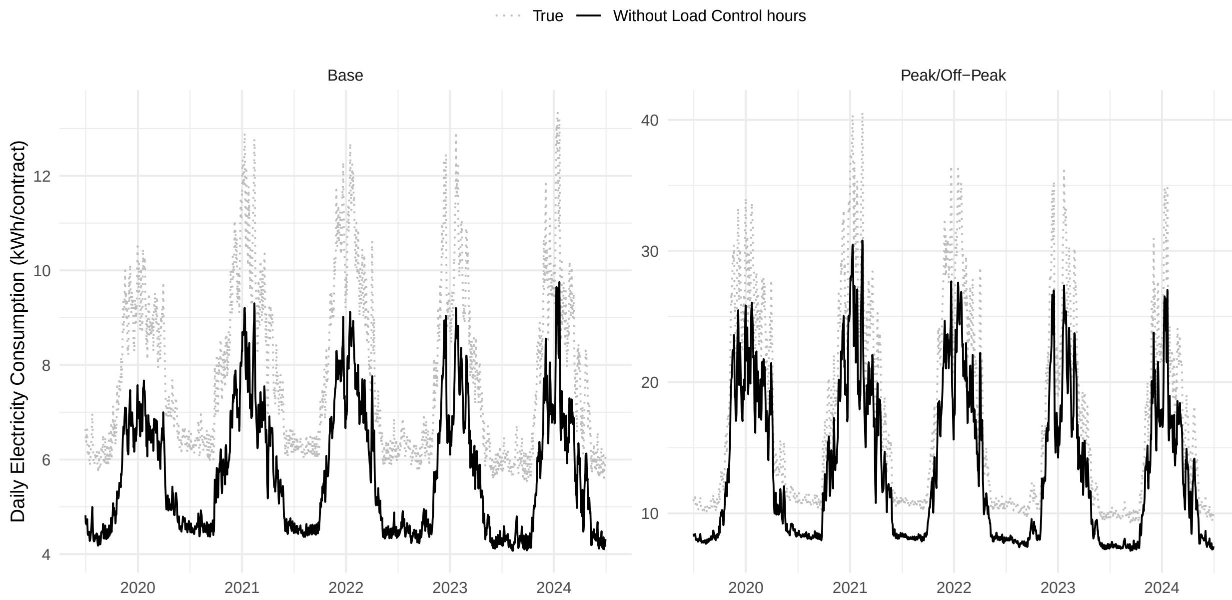

Figure 10: Daily Electricity Consumption with and without Load Control Hours

C.4 Robustness Contracted Power

Table 16: Average Annuel Electricity Consumption - Contracted Power

		Base		ToU		
		6 kVA	9 kVA	6 kVA	9 kVA	12 kVA
Panel A. Household Characteristics						
Housing	Detached Home (%)	40.4	88.9	35.7	75.0	100.0
	Apartment (%)	59.6	11.1	64.3	25.0	0.0
Surface	< 50 m² (%)	14.9	0.0	35.7	0.0	0.0
	≥ 100 m² (%)	19.1	66.7	21.4	66.7	50.0
Income	$< 1,500$ €/month (%)	8.5	0.0	35.7	16.7	25.0
	$\geq 3,500$ €/month (%)	29.8	33.3	14.3	16.7	25.0
Children	0 (%)	83.3	100.0	84.6	53.8	62.5
	2+ (%)	0.0	0.0	0.0	23.1	25.0
End-Use	Heating (%)	16.0	37.8	25.8	34.7	60.6
	Hot Water (%)	42.2	14.4	34.5	30.0	13.5
Sample	Elecdom (N)	47	9	14	12	8
	Enedis (M)	14	3	6	5	3
Panel B. Consumption and Contracts (2019-2021 vs 2022-2023)						
Total Consumption (TWh)						
	2019–2021	33.7	17.4	26.6	38.3	32.8
	2022–2023	31.0	18.9	23.7	33.2	28.1
	Change (%)	-7.9	+8.7	-10.8	-13.4	-14.3
Number of Contracts (Millions)						
	2019–2021	14.4	3.1	6.1	5.2	3.2
	2022–2023	14.5	3.6	6.4	5.2	3.1
	Change (%)	+0.5	+12.8	+3.6	-0.3	-2.0
Average Consumption (MWh/contract)						
	2019–2021	2.3	5.5	4.3	7.4	10.3
	2022–2023	2.1	5.3	3.7	6.4	9.0
	Change (%)	-8.3	-3.4	-13.7	-13.1	-12.6

Notes : Years are defined from July to July in order to encompass one winter period per year of study. Thus, the year 2022 goes from July 2022 to June 2023.

Table 17: Contracted Power - RECM

	Base - 6 kVA		Base - 9 kVA	
Variable	Elast.	Wh/day	Elast.	Wh/day
Panel A. Price and Temperature				
Intercept		$5,153^{***}$		$7,565^{**}$
		(683)		(3,277)
Price	0.1552^{***}	-8.3^{***}	0.2860^{**}	-36.3^{**}
	(0.0232)	(1.2)	(0.1169)	(14.8)
Temperature	0.0282^{*}	42.1^{*}	0.0731	-262.4
	(0.0145)	(21.7)	(0.0910)	(326.5)
Panel B. Attention Effects				
Conservation	0.0022	-26.4	0.0015	-44.4
	(0.0028)	(34.0)	(0.0112)	(331.6)
Crisis	0.0005	-5.6	0.0050^{**}	-128.3^{**}
	(0.0004)	(4.3)	(0.0022)	(56.5)
Observations	1,839		1,839	

Notes : Elasticities represent the ratio of proportional changes in quantity demanded to proportional changes in independent variables; values below 1 in absolute value indicate inelastic demand. Estimates are expressed in Wh/day/contract. Data span July 1, 2019 to June 30, 2024. *** p<0.01, ** p<0.05, * p<0.1.